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Argument-based Person-Tailored Human-Agent Dialogues to Support Health and Well-Being in Older Adults

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Abstract

Supporting health and well-being in later life requires attention to the activities, goals, motives, and contexts that make everyday life meaningful. This thesis investigates how dialogue between a human and a software agent can be person-tailored to support reflection and planning related to health and well-being. The aim is to develop and investigate computational models and mechanisms that relate agent support to a person's activities and motives, dialogue purpose, and context.

The thesis contributes models for representing purposeful human activity and mechanisms for using this knowledge in human-agent collaboration. Grounded in Activity Theory and argumentation frameworks the contributions connect activities with goals, motives, actors, instruments, and context; address different dialogue purposes and potentially competing motives; and support the generation of person-tailored arguments. Using a participatory design methodology, a diary agent was developed in collaboration with occupational therapists; its purpose is to work with the individual to construct diary entries that support reflection on past activities and the planning of future ones. Models and mechanisms were complemented by generative AI, where the generated narratives are guided by argumentation types and argumentation schemes grounded in structured personal information. The interaction design enables collaboration as well as user control and corrections. Argument mining was used to examine the argumentation structure and the extent to which the narratives are based on the individual's information.

Across six formative evaluation studies, the importance was shown of both topic and context for health-related dialogue. Evaluations with domain experts and older adults show opportunities and limitations of person-tailored narratives, including difficulties in communicating intended narrative purposes and the importance of cultural relatability and personal relevance. Technical evaluation shows that argumentation-scheme guidance increases explicit premises but does not independently produce a significant improvement in persona grounding. Argument mining can support inspection of argumentative structure but the study did not provide evidence for supporting autonomous verification. Further, requirements for grounding, explainability, emotional safety, and human ownership were elicited.

To summarise, the thesis shows how purposeful activity, multiple motives,

dialogue purposes, structured reasoning, generative narratives, and human participation can be connected in person-tailored human-agent dialogue while keeping personal meaning and human interpretation central.

Sammanfattning

För att främja hälsa och välbefinnande senare i livet är de aktiviteter, mål, motiv och sammanhang som gör vardagen meningsfull viktiga. Denna avhandling undersöker hur dialog mellan människa och en mjukvaruagent som använder tekniker från området artificiell intelligens (AI) kan individanpassas för att stödja reflektion kring hälsa och välbefinnande och planering av hälsobefrämjande aktiviteter. Syftet är att utveckla och utvärdera beräkningsmodeller och mekanismer som gör att en agents stöd anpassas till en persons aktiviteter och motiv, dialogens syfte samt sammanhanget.

Avhandlingen bidrar med modeller för att representera människors aktivitet så att ett datorprogram kan använda kunskapen, samt mekanismer för en agent att använda denna kunskap i interaktion och samarbete med människan. Med verksamhetsteori och argumentationsteoretiska ramverk som grund kopplar bidragen samman aktiviteter med mål, motiv, aktörer, verktyg och kontext; hanterar olika dialogsyften och potentiellt motstridiga motiv; samt stödjer genereringen av individanpassade argument. Genom deltagande designmetodik utvecklades tillsammans med arbetsterapeuter en dagboksagent som har som uppgift att samarbeta med personen i konstruktionen av dagboksinlägg som stöd till reflektion över tidigare aktiviteter och planering av kommande aktiviteter. Modeller och mekanismer kompletterades med generativ AI där narrativen som genereras styrs av argumentationstyper och argumentationsscheman förankrade i strukturerad personinformation. Interaktionsdesignen möjliggör samarbete och personens kontroll och korrigeringar. En metod baserad på tekniken *argument mining* utvecklades för granskning av argumentationsstrukturen och i vilken mån narrativen baseras på personens information.

I sex studier visar formativ utvärdering vikten av både personligt relevant ämne och kontext för hälsorelaterad dialog. Utvärderingar med områdesexperter och äldre vuxna visar möjligheter och begränsningar med personanpassade berättelser, inklusive svårigheter att kommunicera avsedda narrativa syften och vikten av kulturell relaterbarhet och personlig relevans. Teknisk utvärdering visar att begränsning av språkmodellen med argumentationsscheman ökar mängden explicita premisser som stöd för påståenden, men studien visade inte att det gav signifikant förbättring av personaförankring. En slutsats som drogs var att argument mining kan användas för inspektion av argumentativ struktur

men inte för autonom verifiering av att innehållet är korrekt i förhållande till personens information. Deltagandedesignprocessen identifierade vidare krav på förankring, förklarbarhet, emotionell trygghet och människans ägarskap när människan och mjukvaruagenten samarbetar i skapandet av narrativ om personens tidigare och framtida aktiviteter.

De empiriska studierna ger formativa insikter och designimplikationer av vilka en del implementerades i prototyper för att utvärderas i påföljande studier. Sammantaget visar avhandlingen hur maskinläsbara representationer av målinriktad aktivitet, multipla motiv, syften med dialogen, strukturerat resonerande, skapandet av AI-genererade narrativ modifierade i samarbete mellan agent och människa kan sammanföras i en individanpassad dialog mellan människa och agent om den gemensamma aktiviteten, samtidigt som personlig innebörd och mänsklig tolkning förblir centrala.

Preface

This thesis is based on the following six papers.

Paper I Jayalakshmi Baskar and Helena Lindgren. (2014). **Cognitive Architecture of an Agent for Human-Agent Dialogues.** In: Corchado, J.M., et al. Highlights of Practical Applications of Heterogeneous Multi-Agent Systems. The PAAMS Collection. PAAMS 2014. Communications in Computer and Information Science, vol 430. Springer, Cham. https://doi.org/10.1007/978-3-319-07767-3_9

Paper II Jayalakshmi Baskar and Helena Lindgren. (2015). **Human-Agent Dialogues on Health Topics - An Evaluation Study**. In: Bajo, J., et al. Highlights of Practical Applications of Agents, Multi-Agent Systems, and Sustainability - The PAAMS Collection. PAAMS 2015. Communications in Computer and Information Science, vol 524. Springer, Cham. https://doi.org/10.1007/978-3-319-19033-4_3

Paper III Jayalakshmi Baskar, Chunli Yan, and Helena Lindgren. (2017). **Instrument-Oriented Approach to Detecting and Representing Human Activity for Supporting Executive Functions and Learning**. In Proceedings of the European Conference on Cognitive Ergonomics (ECCE '17) (pp. 105-112). ACM. https://doi.org/10.1145/3121283.3121305

Paper IV Jayalakshmi Baskar, Rebecka Janols, Esteban Guerrero, Juan Carlos Nieves, and Helena Lindgren. (2017). **A Multipurpose Goal Model for Personalised Digital Coaching**. In: Montagna, S., Abreu, P., Giroux, S., Schumacher, M. (eds) Agents and Multi-Agent Systems for Health Care. A2HC AHEALTH 2017 2017. Lecture Notes in Computer Science(), vol 10685. Springer, Cham. https://doi.org/10.1007/978-3-319-70887-4_6

Paper V Jayalakshmi Baskar, Vera C. Kaelin, Kaan Kilic, and Helena Lindgren. (2026) **Argumentation -Based Person-Tailored Storytelling for Older Adults: A Reflective Hybrid AI Archi-**

tecture. arXiv:2605.10531 [cs.AI]. https://doi.org/10.48550/arXiv.2605.10531

Paper VI Jayalakshmi Baskar, Vera C. Kaelin, and Helena Lindgren. (2026) **Understanding human-diary agent collaboration through an activity- theoretical framework**. Manuscript under minor revision, Journal of Ambient Intelligence and Humanized Computing.

Author contributions: Jayalakshmi Baskar was the main contributor to the research reported in the papers included in this thesis. Her contributions included conceptual development, research design and prototype development, empirical work, analysis, and manuscript preparation. The nature and extent of these contributions differed across papers and collaborations.

For Papers I and II, Jayalakshmi Baskar led the design and implementation of the human-agent dialogue system and conducted the reported research in close collaboration with Helena Lindgren. Helena Lindgren contributed to the research direction, conceptual development, supervision, and revision of the manuscripts.

For Paper III, Jayalakshmi Baskar contributed to the conceptualisation, design, and implementation of the instrument-oriented activity representation model, including the event ontology and core ontology. She also contributed to the writing of the paper. Chunli Yan and Helena Lindgren contributed to the conceptual development, research design, supervision, and revision of the manuscript.

For Paper IV, Jayalakshmi Baskar contributed to the development of the multipurpose goal model, the human-agent dialogue component, prototype development, the analysis of the use case, and the writing of the paper. Rebecka Janols, Esteban Guerrero, Juan Carlos Nieves, and Helena Lindgren contributed to the conceptual development, related system components, research design, supervision, and revision of the manuscript.

For Paper V, Jayalakshmi Baskar contributed to the study design, system development, data collection, data analysis, technical evaluation, and manuscript preparation. Vera C. Kaelin, Kaan Kilic, and Helena Lindgren contributed to the conceptual development, study design, interpretation of findings, supervision, and revision of the manuscript.

For Paper VI, Jayalakshmi Baskar contributed to the study design, prototype implementation, data collection, analysis, and manuscript preparation. Vera C. Kaelin and Helena Lindgren contributed to the study design, interpretation of findings, supervision, and manuscript revisions.

In addition to the papers included in this thesis, the following publications were produced during the doctoral studies but are not included in the thesis.

1. Jayalakshmi Baskar, Vera C. Kaelin, and Helena Lindgren. **Experts' Perception of AI-Generated Personalised Stories for Health Pro-**

motion. Studies in Health Technology and Informatics, vol. 336, pp. 660–664, 2026.

2. Jayalakshmi Baskar, Kaan Kilic, Vera C. Kaelin, and Helena Lindgren. **Towards Collaborative Planning for Health Promotion through Person-Tailored Storytelling and Argumentation**. In Proceedings of the 1st Workshop on Human-AI Collaborative Systems, co-located with ECAI 2025, CEUR Workshop Proceedings, vol. 4072, 2025.

3. Helena Lindgren, Jayalakshmi Baskar, Esteban Guerrero, Juan Carlos Nieves, Ingeborg Nilsson, and Chunli Yan. **Computer-Supported Assessment for Tailoring Assistive Technology**. In Proceedings of the 6th ACM International Conference on Digital Health, pp. 1–10, 2016.

4. Juan Carlos Nieves, Esteban Guerrero, Jayalakshmi Baskar, and Helena Lindgren. **Deliberative Argumentation for Smart Environments**. In PRIMA 2014: Principles and Practice of Multi-Agent Systems, pp. 141–149, 2014.

5. Jayalakshmi Baskar, Helena Lindgren, and Chunli Yan. **User's Control of Personalised Intelligent Environments Supporting Health**. In Proceedings of the 9th International Conference on Intelligent Environments, pp. 270–273. IEEE, 2013.

6. Helena Lindgren, Farahnaz Yekeh, Jayalakshmi Baskar, and Chunli Yan. **Agent-Supported Assessment for Personalized Ambient Assisted Living**. In Workshop on Agents Applied in Health Care, pp. 141–150, 2012.

7. Jayalakshmi Baskar, Helena Lindgren, and Chunli Yan. **Personalisation and User Models for Support in Daily Living**. In The 27th Annual Workshop of the Swedish Artificial Intelligence Society, pp. 7–15, 2012.

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nuity and belonging at the department.

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Glossary

ACKTUS A knowledge-based platform used in this research for modelling and managing structured knowledge, including knowledge used in human-agent dialogue.

AIF *Argument Interchange Format.* A formal framework for representing argumentative information and relations between argumentative elements.

AI *Artificial Intelligence.* Computational methods and systems designed to perform tasks involving capabilities such as reasoning, learning, language processing, and decision support.

Argument mining Computational identification of argumentative components, such as claims and premises, in textual content. In this thesis, argument mining is used as an inspection aid for argumentative structure, not as an autonomous verifier of factual correctness or grounding.

Argumentation scheme A recurring pattern of reasoning that structures how premises support a conclusion. Argumentation schemes are used in this thesis to guide argument generation for different dialogue purposes.

BDI *Belief–Desire–Intention.* An agent-modelling framework in which beliefs represent information available to an agent, desires represent goals, and intentions represent committed plans or actions.

Dialogue purpose The communicative goal of an interaction. Dialogue purposes used in this thesis include inquiry, deliberation, persuasion, information-seeking, and support in different parts of the research.

Grounding The extent to which person-specific generated content is supported by information available in the structured person or persona model.

ICF *International Classification of Functioning, Disability and Health.* A World Health Organization framework for describing functioning, disability, health, activities, participation, and contextual factors.

LLM *Large Language Model.* A generative language model trained on large amounts of textual data and capable of generating and processing natural-language content.

Ontology A formal, computer-interpretable representation of concepts and relationships within a domain.

OWL *Web Ontology Language*. A language for formally representing ontologies and relationships between concepts.

Person-tailoring In this thesis, adapting support to information relevant to an individual person, including activities, goals, motives, priorities, experiences, and context.

Purposeful activity Activity understood in relation to the goals, motives, actors, instruments, and context that give the activity meaning and direction.

RDF *Resource Description Framework*. A standard model for representing information as relationships between resources.

Reflective inspection Inspection by an agent of its generated content to make aspects of its structure or relation to represented information available for analysis. It does not imply AI self-awareness or autonomous verification.

User model A structured representation of information about a person that can be used by a computational system to tailor its reasoning or interaction.

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Chapter 1

Introduction

The number and proportion of older adults are rising globally, and this change places new demands on health, social care, and everyday support systems [94, 106]. At the same time, ageing should not be understood only through disease, decline, or service need. The World Health Organization frames healthy ageing as the process of developing and maintaining functional ability, including the ability to meet basic needs, learn, make decisions, be mobile, build relationships, and contribute to society [106, 103]. This perspective shifts attention from illness alone to the conditions that allow people to live meaningful everyday lives.

Meaningful everyday life is not the same for every older adult. Health and well-being are shaped by routines, relationships, activities, environments, values, and personal priorities. Ageing in place, for example, is often valued not only because it allows a person to remain at home, but because familiar places can support autonomy, identity, security, and social belonging [101, 73]. Person-centred and integrated approaches to care similarly emphasise that support should be coordinated around the person rather than around isolated services or single conditions [105, 58]. Support for healthy ageing therefore needs to take seriously what matters to the individual person.

This thesis investigates person-tailored dialogues between a human and a software agent to support health and well-being in older adults. A software agent is a computational system that receives information from its environment and acts upon it [83]; in my thesis, agents are considered particularly in relation to reasoning about human activities and supporting dialogue [Paper I]. In this thesis, *person-tailored* means that the agent's support is grounded in structured knowledge about a particular person rather than directed to an assumed average user. Such knowledge may include the person's activities, goals, motives, preferences, priorities, experiences, dialogue needs, and current context. The central concern is not only whether an agent can produce a relevant utterance, story, or diary text. It is whether the agent can relate its contribution to what the person is doing, why it matters, and what kind of support the

interaction is meant to provide.

The overarching computational research problem is how an artificially intelligent (AI) agent can use structured and situated knowledge about a person to provide support that is purposeful, context-sensitive, and open to human interpretation. This problem consists of several connected challenges. The agent must represent the person and the person's activities, reason about goals and motives in context, relate dialogue moves to the purpose of the interaction, support planning and reflection, and evaluate whether generated content remains grounded in the user model and activity context. These challenges cannot be treated as independent components if the aim is person-tailored support for everyday health and well-being.

Conversational AI and human-agent dialogue offer a relevant opportunity for such support. Conversational agents can ask questions, provide information, help a person examine alternatives, and support reflection or planning outside ordinary clinical encounters [71, 93]. Earlier work on relational and health dialogue agents also suggests that repeated conversational interaction can be acceptable to older adults and may provide an accessible interface for health-related support [15]. Dialogue is important because reflection and planning are not only information-delivery tasks. They involve interpretation, clarification, and response to the person's situation.

However, existing conversational-agent research also shows important limitations. Reviews of health conversational agents describe a field with many task-specific systems, varied implementation strategies, and limited evidence concerning efficacy, safety, and long-term use [57, 93, 71]. Personalisation is increasingly used, but it often concerns selected variables, interface features, content choices, reminders, or behaviour-change strategies rather than a deeper computational account of why an activity matters to a person [55, 90]. Tailoring persuasive eHealth strategies for older adults on the basis of motivation is an important step in this direction, but it also shows that motivation and context are difficult to represent in a way that can guide interaction [95].

The difficulty is that person-tailored support must connect several kinds of information [40]. An activity is not meaningful only because it can be named, detected, or recorded. Walking, preparing food, meeting a friend, resting, or writing a diary entry can each have different meanings depending on the person's goals, motives, constraints, relationships, and situation. The same activity may support health, independence, enjoyment, recovery, social participation, or a sense of purpose. These motives may reinforce one another, but they may also conflict. A system that treats health support as the pursuit of one predefined goal may therefore miss the person's own priorities.

Dialogue adds another layer of complexity. An agent must decide not only what topic to address, but also what the purpose of the interaction is. The agent may ask for information, help the person reflect on a past activity, discuss possible future actions, or encourage the person to consider a health-promoting choice. These purposes require different forms of dialogue and different uses of knowledge. The immediate context of the interaction also matters. A response

may be related to a relevant health topic but still be inappropriate if it does not follow from the person's previous answer or the direction in which the dialogue has developed [Paper I, Paper II].

Person-tailored support requires a user model that represents relevant information about the person and their situation. In this thesis, this includes activities, goals, motives, priorities, dialogue purposes, and context as interconnected aspects of the person's situation. Such a user model should also be transparent to the person and open to inspection and negotiation during interaction [97, 68]. This thesis addresses this challenge through models and mechanisms for representing purposeful activity, organising agent dialogue, and reasoning about multiple motives and goals [Paper III, Paper IV].

These contributions do not by themselves establish clinical effectiveness. Rather, they provide conceptual and computational foundations for asking how an artificial agent can ground its dialogue contributions in structured knowledge about the person and the person's everyday life.

Modern generative AI extends this problem in important ways. Large language models can produce fluent text, flexible responses, and coherent narratives, and they open new possibilities for personalised storytelling, diary writing, reflection, and planning [17, 76]. Digital storytelling research with older adults suggests that stories can support memory, identity, social connection, and self-expression, although evidence for effectiveness remains limited and context-dependent [80, 88]. For this thesis, generative AI is therefore not a replacement for structured person knowledge. Generative AI is therefore a complementary instrument/way to express, communicate, and collaboratively explore and evaluate such knowledge in narrative form.

The ability of generative AI to produce fluent and plausible content also makes grounding more difficult. A fluent story or diary entry may include assumptions that are not supported by the person's information. It may sound coherent while misrepresenting an activity, motive, priority, or cultural context. In health-related settings, these risks are especially important because people may interpret generated content as meaningful feedback about their own lives. Research on hallucination and grounding in large language models, together with health-focused AI governance guidance, emphasises the need for reliability, transparency, human oversight, and careful evaluation [46, 42, 104, 51].

These concerns are also relevant to responsible AI development, where AI systems should be designed and evaluated with attention to reliability, transparency, human oversight, and potential risks. More broadly, AI alignment concerns the development of AI systems whose behaviour is consistent with human intentions, goals, and values [46, 86]. In this thesis, alignment is used in a narrow sense: consistency between the user model, the intended dialogue purpose, the activity context, and the generated dialogue or narrative content.

These concerns extend the original grounding problem. The agent must not only generate language, but also support reflective analysis of generated content in relation to the structured knowledge and dialogue purpose that guided it [Paper V]. Such analysis should not be understood as AI self-awareness or as a

guarantee that the content is correct. It is a mechanism for inspecting whether generated dialogue or narrative content appears consistent with the modelled person information, argument-based purpose, and activity context.

Person-tailored support also requires human participation. A completed AI-generated narrative can leave the person only as a reader or evaluator of the agent's output. Reflection and planning often require more than that. The person may want to correct an assumption, reject an interpretation, add missing context, question the purpose of the agent's suggestion, or reshape the narrative in their own words. Human-AI interaction and collaboration research therefore emphasises agency, transparency, appropriate control, and opportunities for correction and refinement [3, 87, 31, 100]. In the later work of this thesis, these issues are examined through expert-informed design of collaborative diary interaction, where the diary text is treated as a shared artefact developed through dialogue rather than as a fully validated intervention [Paper VI].

Previous work has typically addressed these capabilities separately. Research on conversational agents, eHealth tailoring, digital storytelling, generative AI, and human-AI collaboration provides important parts of the landscape. However, relatively little work has integrated structured knowledge about people, purposeful everyday activities, multiple motives, dialogue purposes, context, generative narrative, reflective analysis of generated content, and collaborative human participation in a unified person-tailored dialogue framework for supporting both planning and reflection in older adults. Developing such integrated approaches remains an open challenge, which is a research gap and challenge addressed in this thesis.

The overall aim of the thesis is to investigate how person-tailored human-agent dialogues can be modelled, implemented, and evaluated to support health-related reflection and planning with a focus on older adults. Across the six included papers, the thesis develops this aim through conceptual foundations, computational models, prototyping, expert participation in design and evaluation, design recommendations, empirical studies, and iterative refinement [Paper I, Paper II, Paper III, Paper IV, Paper V, Paper VI]. The thesis does not claim to provide a completed clinical intervention or a deployed production system. It examines how structured knowledge, dialogue purposes, generative narratives, reflective analysis, and human participation can be brought together as foundations for person-tailored support for health and well-being.

1.1 Research Problem

The research problem addressed in this thesis is how an artificial agent can represent and use knowledge about a person, the person's activities, motives, goals, priorities, dialogue purposes, and context to support and collaborate with the person through dialogues with the person that are person-tailored, purposeful, and open to human interpretation and revision.

Addressing this problem requires connecting several computational and

human-centred concerns. These include representing purposeful activities and their context, reasoning about multiple motives and goals, relating agent contributions to different dialogue purposes, generating person-tailored content while maintaining grounding in represented information, and enabling generated content to be inspected, interpreted, and revised. The thesis investigates these concerns through participatory design involving domain experts, grounding in theories on human activity and reasoning, complementary models and hybrid AI, mechanisms, prototypes, and empirical studies rather than through a single fully integrated intervention.

1.2 Research Questions

The research questions are:

RQ1 How can knowledge about users, activities, goals, motives, dialogue purposes, and context be represented to support person-tailored human-agent dialogues?

RQ2 How can cognitive agent architectures integrate such knowledge to support purposeful reasoning and interaction in person-tailored human-agent dialogues?

RQ3 How can person-tailored agents support health-related reflection and planning through argumentation, storytelling, and reflective inspection?

RQ4 How do domain experts and older adults perceive and experience the support provided by such agents?

1.3 Research Contributions

The main contribution of this thesis is a set of models, mechanisms, and empirical insights for person-tailored human-agent dialogue that connect purposeful human activity with dialogue, reasoning, and reflection. The thesis develops this contribution through four related areas corresponding to the research questions.

1. Computational representation of purposeful activity for person-tailored dialogue. The thesis contributes models for representing information about a person together with activities, goals, motives, priorities, dialogue purposes, and context. A central contribution is the representation of purposeful human activity in a form that connects observed or reported events with actors, activities, instruments, goals, motives, and contextual information. This provides a basis for moving from information about what a person does towards computational representations of why an activity may matter in a particular situation. Paper III develops and implements the activity ontologies underlying this contribution, while Papers I and IV provide related representations of user knowledge, dialogue context, motives, and goals.

2. Models and mechanisms for purposeful and multipurpose human-agent reasoning. The thesis contributes an architectural and reasoning approach in which person-tailored interaction is organised according to both the person's represented situation and the purpose of the dialogue. The proposed cognitive architecture separates beliefs, dialogue management, task management, and planning, with partial implementation of the dialogue-management mechanisms. The thesis further develops an early approach for representing multiple potentially complementary or conflicting motives and complementary agent roles. Argumentation provides mechanisms for selecting and combining reasons related to these motives rather than assuming a single predefined health objective. Papers I and IV provide the main evidence for this contribution, with Paper II providing formative evidence concerning contextual appropriateness in health-related dialogue.

3. Argumentation-guided generation and inspection of person-tailored narratives. The thesis contributes a hybrid approach that combines structured person information, dialogue purposes, argumentation schemes, and large language models for generating person-tailored narratives for reflection and planning. Argumentation schemes provide explicit reasoning guidance during generation, while structured persona information provides a reference for assessing whether person-specific statements are grounded. The technical evaluation shows that scheme guidance increases the expression of explicit premises, but does not establish a statistically significant independent improvement in persona grounding. The thesis also evaluates argument mining as a mechanism for inspecting argumentative structure in generated narratives. Argument mining is used as an inspection aid and not as an autonomous verifier of factual correctness, grounding, or hallucination. Paper V provides the main evidence for this contribution.

4. Empirical findings supporting new design implications for person-tailored reflective interaction. The thesis contributes empirical and design knowledge concerning how person-tailored dialogue and narrative support are perceived and how collaborative interaction can be designed. Formative evaluation in Paper II shows the importance of considering the immediate dialogue context in addition to topic relevance. Paper V provides evidence from domain experts and older adults concerning generated narratives, including the finding that computationally specified dialogue purposes are not necessarily perceived as intended by the participants. Paper VI extends the interaction perspective through participatory design of human-diary agent collaboration and identifies requirements concerning grounding, explanation, ownership, flexibility, emotional safety, and opportunities for human interpretation and revision. These contributions provide design knowledge rather than evidence of clinical effectiveness or long-term health outcomes.

Together, the four contributions trace a progression from representing purposeful activity and person-specific context, through modelling dialogue purposes and multiple motives, to argumentation-guided narrative generation and reflective inspection, and finally to collaborative interpretation and revision

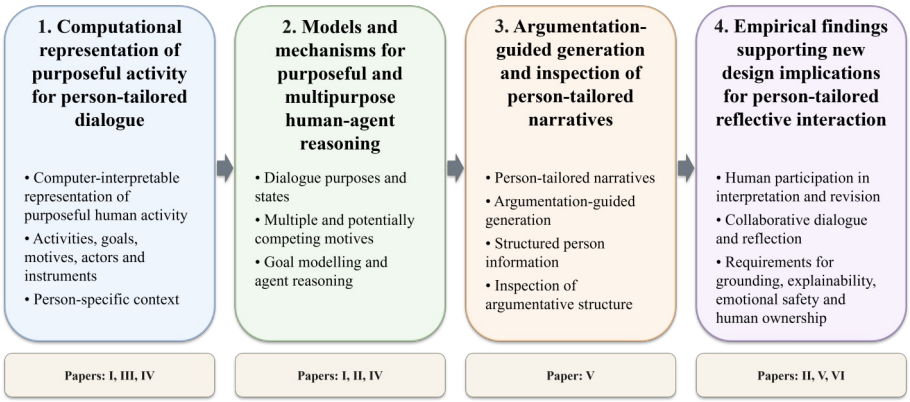


Figure 1.1: Overview of the four thesis contributions and their relationship to Papers I-VI. The contributions represent complementary developments across the thesis.

(refer Figure 1.1). The contributions should not be understood as a single fully implemented and clinically evaluated system. Rather, they consist of complementary conceptual models, implemented representations and mechanisms, research prototypes, empirical findings, and design requirements developed across the six papers.

1.4 Scope and Delimitations

This thesis focuses on computational and human-centred approaches to person-tailored human-agent dialogue for health-related reflection and planning, with older adults as the main target group. The research examines how information about activities, goals, motives, priorities, dialogue purposes, and context can be represented and used in dialogue, argumentation, narrative generation, and collaborative reflection. The scope includes conceptual models, knowledge representations, agent architectures, implemented mechanisms and research prototypes, together with formative, participatory, and technical evaluations.

The thesis does not develop or evaluate a single integrated system containing all of the mechanisms investigated across the six papers. The papers represent different stages and aspects of the research. Some contributions are conceptual or representational, while others are partially implemented or explored through research prototypes. Consequently, findings from one prototype or study are not treated as evidence that all proposed components have been implemented or evaluated together.

The empirical scope is also delimited by the participant groups and study designs. Paper II reports a formative evaluation involving five therapists and eleven older adults. Paper V includes a formative study with eleven domain

experts and an evaluation with 55 older adults. In the latter study, the older adults evaluated narratives generated from selected fictional personas rather than narratives based on their own personal histories. In Paper VI participatory design was conducted with six occupational-therapy experts and an exploratory prototype; it does not evaluate outcomes with older adults. These studies provide complementary evidence about interaction, perception, and design, but they do not constitute a longitudinal or clinical evaluation of one intervention engaging the target group.

The technical evaluations are similarly bounded. In Paper V, persona grounding is assessed separately by comparing generated person-specific statements with structured persona information. Argument mining is evaluated as a method for identifying argumentative functions in generated text. It is not used as an autonomous method for verifying factual correctness, detecting hallucinations, or determining whether statements are grounded. The technical evaluation therefore concerns specific properties of generated narratives rather than the general reliability or safety of large language models.

Finally, the thesis does not evaluate clinical effectiveness, behaviour change, or long-term health and well-being outcomes. It does not claim that the investigated agents replace healthcare professionals or human decision-making. Instead, the thesis contributes foundations, mechanisms, empirical findings, and design requirements for computational systems intended to support person-tailored reflection, planning, and human-agent interaction. Questions concerning longitudinal use, clinical effectiveness, deployment, and the evaluation of fully integrated systems remain outside the scope of the present thesis.

1.5 Thesis Outline

The remainder of this thesis is organised as follows. Chapter 2 introduces the theoretical and technical background for the research, including Activity Theory, human-agent dialogue, person-tailoring, cognitive architectures, knowledge representation, argumentation, and generative AI with grounding and reflective inspection by an agent.

Chapter 3 presents the modelling foundations for person-tailored human-agent dialogue. It describes how purposeful human activity is connected to dialogue support, how users, activities, motives, goals, and context are represented, and how dialogue purposes and agent roles are incorporated into the proposed models and architectures.

Chapter 4 describes mechanisms for supporting collaborative reasoning and dialogue. It examines dialogue as a collaborative activity, the relationship between dialogue types and argumentation schemes, person-tailored narrative generation and reflective inspection, and the progression from conceptual models to implemented mechanisms and research prototypes.

Chapter 5 describes the research methodology and evidence base of the thesis. It distinguishes conceptual and technical development from formative eval-

uation, participatory design, evaluation with older adults, and computational assessment. It also presents the technical evaluation of argumentation-based narratives and summarises findings across the studies.

Chapter 6 presents the six papers included in the thesis and summarises their individual contributions and relationship to the overall research. Chapter 7 discusses the findings at the thesis level in relation to the research questions, contributions, limitations, and broader implications. Chapter 8 concludes the thesis by summarising its main contributions and conclusions and identifying directions for future research.

The six included papers follow the kappa.

Chapter 2

Background on Theoretical Frameworks

This chapter introduces the theoretical and technical concepts that provide the foundation for the thesis. The research concerns human-agent interaction in which support needs to take account of what a person does, why activities matter to that person, the purpose of the interaction, and the context in which it occurs. Addressing these aspects requires concepts from several areas rather than a single theoretical framework.

Activity Theory provides a perspective on purposeful human activity and the relationship between activities, motives, goals, actors, and mediating instruments. Human-agent dialogue provides the interactive setting in which people and computational agents exchange information and reason about topics of interest. Person-tailoring concerns how such interaction can be adapted using information about the individual and the situation. Cognitive architectures provide structures for organising the knowledge and processes required by an agent, while knowledge representation provides computer-interpretable forms for information about activities, people, and context. Argumentation provides mechanisms for representing reasons, claims, and different purposes of dialogue. Finally, generative AI introduces new possibilities for producing person-tailored narratives, together with challenges concerning grounding and the inspection of generated content.

These foundations are described separately in this chapter for clarity. Chapters 3 and 4 show how they are brought together in the models and mechanisms developed in the thesis.

2.1 Activity Theory

Activity Theory provides a framework for understanding human activity as purposeful and situated rather than as a sequence of isolated actions. Its ori-

gins lie in the cultural-historical tradition associated with Vygotsky and were subsequently developed through the work of Leont'ev and others [96, 59]. A central idea is that human activity is directed towards an object and motivated by needs or motives. Activities are realised through actions directed towards goals, while actions can in turn be carried out through operations that depend on the conditions of the situation [59].

Another important concept is **mediation**. People generally do not act directly on the world without means of support. Activity is mediated by instruments, which can include physical tools as well as symbolic and cognitive resources [96, 49]. The relationship between the person, the purpose of an activity, the instruments used, and the surrounding conditions is therefore relevant when attempting to understand why and how an activity is performed.

This perspective is useful for assistive and health-related technologies because observing that an activity occurred does not necessarily explain its meaning to the person. The same observable activity can serve different purposes. For example, taking a walk may contribute to physical exercise, social participation, enjoyment, recovery, or several motives at the same time. Conversely, similar motives can be fulfilled through different activities. Representing activity for person-tailored support therefore requires information beyond observable behaviour.

Activity Theory also provides a basis for considering technologies themselves as mediating instruments [49]. A digital agent may support a person in reflecting on an activity, planning an activity, or considering alternatives, but the agent participates in an activity that remains connected to the person's purposes and circumstances [48]. This perspective is particularly relevant to this thesis because the research moves from representations of purposeful activity towards computational support for reasoning and dialogue concerning those activities.

Activity Theory has also been applied in human-computer interaction and interaction design to understand technology use as part of purposeful and socially situated human activity [49]. Engeström's development of the activity-theoretical perspective further emphasises activity systems that involve mediating artefacts, rules, division of labour, and social relations, and how such systems develop and transform over time [32]. These perspectives provide a basis for examining technology not as an isolated tool, but in relation to the activities, purposes, and contexts in which it is used.

The thesis does not attempt to implement Activity Theory in its entirety. Instead, concepts including **activity, actor, goal, motive, context, and instrument** provide a theoretical basis for the representations and models developed later.

In my research, I focus on the use of these concepts at different stages of human-agent interaction. Paper III uses an activity-centred perspective to represent purposeful human activity computationally, with particular attention to the actor, goals, context, and the instruments mediating activity [Paper III]. Paper VI extends the activity-theoretical perspective to human-agent collab-

oration, where constructing and revising a diary post is conceptualised as a mediated activity involving the person and the diary agent [Paper VI]. Across these studies, Activity Theory thus provides a conceptual link between representing purposeful human activity and examining purposeful collaborative activity with an agent.

2.2 Human-Agent Dialogues

Human-agent dialogue refers in this thesis to interaction between a person and an artificial software agent through which information, questions, interpretations, reasons, or possible actions can be exchanged. Dialogue is relevant to health-related support because interaction may involve different purposes.

Conversational and relational agents have been investigated for health-related interaction, including dialogue with older adults and health counselling [15, 14]. A person may seek information, investigate a situation, consider what to do, evaluate alternatives, or reflect on previous activities.

A useful distinction can therefore be made between the **topic** of a dialogue and its **purpose**. Two dialogues may concern the same health-related topic while pursuing different goals. For example, a dialogue concerning physical activity may seek information about what has happened, inquire into reasons for a situation, deliberate about future activities, or attempt to persuade a person to reconsider an action or priority. Dialogue support consequently requires more than selecting content associated with a topic. Dialogue-management research has addressed this problem through finite-state, frame-based, agent-based, and combined dialogue strategies [70]

Dialogue is also contextual. The appropriateness of an agent contribution depends on what has already occurred in the interaction, what information is available, and what the participants are trying to achieve. An utterance can be relevant to the general topic while being unsuitable at a particular point in the conversation. This distinction between *topic-related* and *contextual appropriateness* became important in the formative evaluation reported in Paper II.

The research in this thesis draws particularly on classifications of argumentation-based dialogue in which different dialogue types are characterised by different communicative goals. These include **information-seeking, inquiry, deliberation, and persuasion** [98]. Earlier work in the thesis also includes support dialogue [Paper IV]. The dialogue types provide a way of making the purpose of interaction explicit and of connecting that purpose to the actions available to an agent.

The meaning of human-agent dialogue becomes broader in the later studies. In Paper V, the immediate output evaluated by older adults is a generated narrative rather than a live conversational exchange. In Paper VI, diary content becomes a shared artefact around which a person and an agent can interact. In this thesis, these narratives are therefore treated as artefacts within the wider

setting of purposeful human-agent interaction rather than assuming that every generated narrative is itself a dialogue.

2.3 Person-Tailoring

Person-tailoring refers to adapting computational support using information that is relevant to a particular person and situation. Personalisation can occur at different levels, ranging from relatively simple adaptation of presentation or content to approaches that use information about a person's activities, preferences, goals, context, or previous interaction [45].

The term **person-tailored** is used throughout this thesis to emphasise that the relevant information extends beyond demographic characteristics or a static user profile. In the context of this research, person-tailoring can include activities, goals, motives, priorities, preferences, contextual circumstances, and information contributed during interaction. The purpose is not to construct a complete computational representation of a person. Instead, the representation contains information considered relevant to the particular support being provided.

User modelling [19, 97, 54] provides a computational basis for this process. A user model represents selected information about the person in a form that a system can use when selecting, generating, or adapting support. Such information may be provided explicitly by the person, obtained through interaction, inferred from observations, or represented through structured domain knowledge. The reliability and meaning of these sources can differ, making it important to retain the connection between represented information and its context. Research on scrutable user modelling has further emphasised the importance of allowing users to inspect and understand the information used for adaptation [50]. This perspective is relevant to person-tailored dialogue because represented information about activities, goals, and preferences may be incomplete or may require clarification through dialogue.

Person-tailoring is particularly important when activities can have several meanings or motives. A recommendation based only on a generic health objective may conflict with what a person values or with other priorities in everyday life. Tailoring can therefore involve not only selecting an activity but also selecting reasons for discussing it and deciding what form of support is appropriate.

The models developed in this thesis progressively extend the information available for person-tailoring. Earlier work represents user information and dialogue context, while Papers III and IV make activities, instruments, goals, motives, and priorities more explicit. Later work uses structured activity and motivation profiles as grounding for generated narratives. This progression should not be interpreted as the development of a complete model of a person. Rather, it represents selected aspects needed for the particular dialogue and reflection tasks investigated in the thesis.

2.4 Cognitive Architectures

A cognitive architecture describes an organisation of components and processes intended to support intelligent behaviour. In agent-based systems, an architecture can specify how an agent represents knowledge, selects goals, constructs plans, manages interaction, and responds to changes in its environment or dialogue context [102].

One influential approach to modelling rational agents is the **Belief-Desire-Intention (BDI)** framework [20, 79]. Beliefs represent information available to an agent about itself, other actors, or the environment. Desires represent states or goals that the agent may seek to achieve. Intentions represent goals or plans to which the agent has committed. The distinction provides a conceptual structure for connecting knowledge with goal-directed action. In healthcare, Fox et al., developed a canonical agent model that integrates concepts including beliefs, goals, planning, decision-making, action, argumentation, and learning [36].

For human-agent interaction, cognitive architectures must also account for dialogue. An agent needs mechanisms for maintaining information relevant to an interaction, identifying the purpose and state of the dialogue, selecting an appropriate next action, and potentially revising a plan when new information becomes available. This creates a relationship between the agent's internal reasoning and the observable dialogue.

Multi-agent perspectives provide another relevant concept. Different agents can be associated with different roles, responsibilities, goals, or domains of knowledge. Such separation can be useful when support concerns several aspects of a person's situation. However, the goals associated with different roles may also conflict, creating a need for coordination and prioritisation.

These concepts provide background for the architecture investigated in Papers I and IV. Paper I proposes a cognitive architecture containing components concerned with beliefs, dialogue management, task management, and planning. Importantly, the complete proposed architecture was **not fully implemented**; the dialogue-management components were developed further while other components remained part of the architectural proposal. Paper IV later explores an early approach in which complementary agent roles and multiple motives are represented within person-tailored digital coaching. The paper describes the agent model in terms of beliefs, desires, intentions, roles, responsibilities, and knowledge, and implements parts of this approach in the NejaVis dialogue system [Paper IV].

2.5 Knowledge Representation

Knowledge representation concerns how information about a domain is expressed in a form that a computational system can store, relate, query, and use for reasoning. In person-tailored human-agent interaction, relevant knowledge

can include information about the person, activities, other actors, environmental circumstances, goals and motives, dialogue history, and domain-specific health knowledge.

Ontologies provide one approach to representing this information. An ontology specifies concepts in a domain and relationships among them, providing a shared conceptual structure that can be interpreted computationally [37]. Semantic representations can make relationships explicit rather than treating information as unrelated data fields. This is useful when an agent needs to connect, for example, an observed event with an activity, the person performing it, an instrument involved in the activity, and a goal or motive.

The Resource Description Framework (RDF) and Web Ontology Language (OWL) provide standards for expressing semantic information and ontologies [23, 22]. Knowledge represented in these forms can be organised as interconnected entities and relations, commonly described in contemporary systems as knowledge graphs. Such representations provide a structured source against which computational reasoning or generated content can be related.

The thesis also draws on health-related conceptual structures. The **International Classification of Functioning, Disability and Health (ICF)** provides concepts concerning body functions and structures, activity and participation, and environmental factors [74]. The ICF has also been used to structure and standardise information about functioning in electronic health records and other health-information systems [69]. Such classifications can provide shared terminology for organising health-related knowledge, although they do not by themselves provide a complete representation of a person's everyday purposes and motives.

The ACKTUS platform used in several of the included studies provides ontology-based knowledge representation for knowledge-based applications [65]. Paper III extends this direction through complementary event and core ontologies for representing human activity. Its model connects detected events and observations with activities and other information needed for person-tailored support. The paper describes this representation as a basis for later personalisation and explicitly leaves user evaluation as future work [Paper III].

Structured knowledge becomes particularly relevant again with generative AI. A generative model can produce fluent content without an explicit structured representation of the person. Providing structured information about activities, preferences, motives, and priorities creates an external reference against which person-specific generated statements can be constrained or subsequently assessed. This relationship between knowledge representation and generation is developed further in the later parts of the thesis.

2.6 Argumentation

Argumentation concerns the construction and evaluation of claims together with the reasons that support or challenge them. It is relevant to human-

agent dialogue because health-related interaction often involves uncertainty, alternatives, competing considerations, and situations in which more than one course of action can be reasonable.

Argumentation has been studied from different theoretical perspectives. Toulmin's model describes argument in terms of elements such as claims, data, warrants, backing, qualifiers, and rebuttals [92]. while Walton's work examines recurring patterns of reasoning and the purposes for which arguments occur in dialogue [98, 99, 81]. Formal argumentation frameworks can represent conflicting arguments and determine which arguments can be accepted under specified relations and semantics [30]. Such approaches have been extended to computational dialogue and multi-agent approaches where agents can construct, exchange, challenge, and respond to arguments according to the purpose and rules of the dialogue [16]. Individual dialogue types have also been studied computationally through dialogue-game models and implemented systems, for example for inquiry dialogue [16].

Argumentation schemes describe recurring patterns of reasoning [99]. A scheme provides a structure connecting premises to a conclusion and may be associated with questions concerning when the reasoning should be accepted. Examples relevant to the research in this thesis include reasoning from values, expert opinion, position to know, and sufficient conditions. Argumentation schemes can therefore connect the communicative purpose of an interaction with the type of reasons presented.

The **Argument Interchange Format (AIF)** is an ontology-based framework designed to support the representation and exchange of argumentative information on the Web [21]. It distinguishes information nodes from scheme nodes and supports explicit representation of relationships among information and inference structures. An adapted version of the AIF ontology is embedded in the ACKTUS core ontology underlying several papers in this thesis.

Argumentation-based approaches have been investigated in digital coaching and explanation systems, including health-related applications [43, 62, 25, 53]. However, studies of persuasive dialogue also indicate that the way arguments are communicated matters; argumentation and advice-giving may elicit resistance, and more client-centred interaction styles may sometimes be perceived more positively [72]. Related work has also explored connections between formal argumentation and Activity Theory for motivation-driven intelligent interfaces [38]. Such approaches provide mechanisms for representing reasons, alternatives, and potentially conflicting considerations and for selecting arguments according to a participant's concerns and preferences [43, 63, 4, 39].

Argumentation also connects with dialogue types. Inquiry can concern establishing or examining information, deliberation concerns what action to take, and persuasion concerns changing or supporting a position or priority. The appropriate form of argument therefore depends partly on the purpose of the dialogue. Paper IV, for example, develops mechanisms for combining reasons related to different motives into multipurpose arguments and describes strategies for selecting arguments in person-tailored coaching [Paper IV].

In the later storytelling work, argumentation schemes are used to guide narrative generation. This should be distinguished from **argument mining**, which concerns identifying argumentative units or relations in text. Recent work has increasingly investigated the use of LLMs for argument-mining tasks, including claim and premise identification, relation analysis, and other forms of argument evaluation [61]. Related work has also investigated whether LLMs can identify argumentation schemes in natural-language arguments, showing promising performance for larger models while also highlighting challenges arising from variation and ambiguity in naturally expressed arguments [13]. In Paper V, argument mining is used to inspect whether generated narratives contain units functioning as claims, premises, or non-argumentative content. It does not establish whether the statements are factually correct or grounded in information about the person. Those properties require separate evaluation. The latest Paper V explicitly treats argument mining and persona grounding as different analyses [Paper V].

2.7 Generative AI, Grounding, and Reflective Inspection

Generative AI, particularly large language models (LLMs), enables systems to generate fluent text in response to instructions and contextual information [17]. For person-tailored interaction, this creates possibilities that differ from earlier dialogue systems based mainly on predefined content and explicit dialogue rules. An LLM can transform structured information into narratives, summaries, or conversational responses and can vary the form of an output according to an intended communicative purpose.

This flexibility also creates a problem. Fluent generated text is not necessarily supported by the information supplied to the model [46, 56]. A generated narrative may introduce assumptions, details, motives, or interpretations that are absent from the person's represented information. Related approaches, such as retrieval-augmented generation, have investigated grounding language-model outputs in external information sources [60]. In this thesis, structured activity and motivation information provides a person-specific reference for guiding narrative generation and separately assessing the grounding of generated statements. In a person-tailored system, **grounding** therefore concerns whether person-specific generated statements are supported by the information that is available about the person.

Grounding should be distinguished from argumentative structure. A statement may be supported by a person's represented information without containing an explicit argumentative premise. Conversely, a narrative can contain a clearly expressed premise supporting a claim even when the premise itself introduces information not supported by the person's profile. Structural argumentation and grounding consequently require different forms of assessment.

Reflective inspection refers here to mechanisms that make aspects of gen-

erated content available for analysis rather than treating the generated text as an unquestioned final output. This can include identifying claims and premises, comparing generated person-specific statements with structured information, and enabling people to inspect, question, or revise generated interpretations. The term does not imply that the AI system possesses human-like self-awareness or that it can autonomously determine the truth of its own output.

Paper V investigates this distinction through a hybrid architecture combining structured user information, dialogue purposes, argumentation schemes, LLM-based narrative generation, and subsequent analysis. Persona grounding is assessed separately against structured persona information, while argument mining is evaluated against human-consensus annotations of CLAIM, PREMISE, and NONE units. The study explicitly states that argument-mining performance is not evidence of persona grounding, factual correctness, hallucination detection, or general AI alignment [Paper V].

The later collaborative diary work adds a human-centred dimension to this issue. Generated content can become an object for collaborative interpretation rather than a completed account that the person is expected simply to accept. This makes grounding and inspection relevant not only as technical properties but also as part of maintaining human participation in the interpretation and revision of person-tailored content.

Together, the concepts introduced in this chapter provide the foundation for the following chapters. Activity Theory supplies concepts for understanding purposeful activity; person-tailoring and knowledge representation concern what information about a person and context can be represented; cognitive architectures concern how an agent can organise and use such information; human-agent dialogue and argumentation concern the purposes and reasoning processes through which support is provided; and generative AI extends these mechanisms towards narrative generation while introducing additional requirements for grounding and inspection. Chapter 3 describes how these foundations are used to model person-tailored human-agent dialogue in the research presented in this thesis.

Chapter 3

Modelling Person-Tailored Human-Agent Dialogues

Person-tailored human-agent dialogue requires a computational representation of both the person and the purpose of the interaction [Paper IV] [1, 98, 66]. Information that an activity has occurred is not sufficient on its own. The agent may also need to represent what the activity means to the person, which goals and motives are associated with it, what contextual information is available, and what purpose the dialogue is intended to serve [Paper IV].

The research in this thesis develops these representational requirements at different levels. It moves from an architecture for purposeful health-related dialogue towards richer representations of human activity and models of multiple motives and agent roles. These models later provide foundations for argumentation-guided narrative generation and collaborative reflection. The development is not a single fully implemented architecture. Rather, the papers contribute complementary conceptual models, implemented representations, and prototype mechanisms for supporting person-tailored human-agent dialogue about health-related activities.

This chapter synthesises these contributions around the information and reasoning required for person-tailored dialogue. Section 3.1 connects purposeful human activity with dialogue support. Section 3.2 describes the representation of people, activities, motives, goals, and context. Section 3.3 considers dialogue purposes and complementary agent roles. Section 3.4 describes the cognitive architecture through which these forms of information can be used in purposeful dialogue.

3.1 Related Work on Modelling Person-Tailored Human-Agent Dialogue

As discussed in Chapter 2, person-tailoring in human-agent interaction can draw on user models, contextual information, knowledge representations, and dialogue models. Rather than repeating this broader background, this section focuses on how these modelling approaches relate to the representational problem addressed in this thesis.

Related approaches differ in what information they represent for tailoring support. Recommender systems commonly construct user profiles from previous behaviour to select relevant content, while assistive systems may tailor support based on individual characteristics, activities, location, or environmental constraints [1, 66, 75]. Agent-based approaches have also been developed for different support purposes, including activity support, monitoring, and health-related decision support [91, 24, 89]. Recent work with older adults has also explored adaptive conversational support in situated activities, for example by tailoring cooking guidance to users' competence and autonomy preferences while combining task-oriented and social interaction [35]. However, representing purposeful activity also requires relating observable information to the actor, the context in which an activity occurs, and the goals and motives that give the activity meaning.

The approach developed in this thesis builds on these directions by representing person-tailoring at several connected levels. Paper III relates observations of activity to actors, instruments, context, goals, and motives, while Paper IV extends the modelling towards multiple, potentially conflicting motives and complementary agent roles. Together with the dialogue architecture developed in Paper I, these contributions provide complementary representations for reasoning about what support is relevant, why it is relevant to the person, and for what purpose the agent engages in dialogue.

3.2 From Identifying Activity To Provide Support Through Dialogue

The starting point for person-tailored support in this thesis is **purposeful human activity**. An observed event, such as walking, meeting another person, or performing an everyday task, provides only part of the information needed to understand the activity. The same observable action may serve different purposes for different people, or several purposes for the same person. Its meaning can depend on the person's goals and motives, the instruments used, other people involved, and the surrounding situation [64].

This creates a gap between **detecting an event** and **interpreting an activity for support**. Sensor data or other observations may indicate that something happened [78, 52, 26]. However, person-tailored dialogue may re-

quire additional information about why it happened and how it relates to what matters to the person. Recent research has highlighted the importance of context awareness and personalisation in activity recognition [29], while goal-recognition approaches [2] investigate how possible goals can be inferred from observations . These approaches provide important foundations for activity understanding. Building on these directions, this thesis investigates how observed activities can be related to a person's goals, motives, and context, and how this information can be represented and used in person-tailored human-agent dialogue.

Paper III addresses this problem by developing a computer-interpretable model that connects events and observations with actors, activities, instruments, goals, motives, and contextual information. The model is implemented through complementary ontologies and provides a common representation for events generated when a person conducts an activity [Paper III].

The representation is influenced by Activity Theory, introduced in Chapter 2. From this perspective, an activity is directed by a motive, while goals organise purposeful actions and instruments mediate activity. This makes the representation useful for distinguishing the physical occurrence of an action from its significance within the person's everyday life. The model does not attempt to represent every aspect of human activity. It provides selected computational structures required for interpreting activity in the applications considered in the thesis.

Paper III also distinguishes activities according to their dependence on domain knowledge, social knowledge, and the physical environment. This distinction reflects that different forms of activity may require different sources of information for detection and interpretation. The representation can therefore combine information obtained computationally with information supplied through human-agent interaction. Paper III specifically identifies dialogue as one possible way of obtaining the person's subjective account of everyday life and complementing information obtained from activity-recognition systems [Paper III].

Dialogue consequently has two related functions within this modelling approach. It can **use represented activity information** to provide person-tailored support, and it can **contribute new information** that extends the representation of the person's situation. A person may explain a preference, clarify why an activity was important, provide information that cannot be observed automatically, or reconsider a previous goal. The user model is therefore not necessarily a fixed profile established before interaction.

This relationship between activity and dialogue becomes important throughout the thesis. Earlier work focuses on information-seeking and health-related dialogue. The multipurpose model later connects activities to several motives and forms of support. In the storytelling work, structured activities and motivations become part of the information used to ground generated narratives. The underlying principle remains that person-tailored interaction requires a connection between observable or reported activity and its purpose in the per-

son's situation.

3.3 Representing Users, Activities, Motives, Goals, and Context

A person-tailored agent requires a user model containing information relevant to the interaction. In the early architecture developed in Paper I, a user model and a domain model are constructed using information relevant to the selected dialogue topic. Missing information in the user model can then become a reason for the agent to ask questions. Information obtained during dialogue can in turn become part of the knowledge available for subsequent reasoning [Paper I].

Paper III develops the activity-related part of this representation in greater detail. Rather than representing only characteristics of a person, it models relationships among the **actor, activity, event, instrument, goal, motive, and context**. Events provide records associated with the performance of activities, while the broader model provides information needed to interpret those events. This creates a computer-interpretable connection between what has been observed and the purposeful activity in which it occurs [Paper III].

A further requirement is that one activity may relate to **multiple motives**. Paper IV addresses this issue through a multipurpose motivational and goal model for digital coaching. The model is based on the observation that support directed towards one health goal may overlook other reasons why a person performs or avoids an activity. The paper therefore represents several potentially complementary or conflicting motives rather than assuming one fixed objective for all support [Paper IV].

In the prototype described in Paper IV, the initial motivational model is populated through a baseline assessment. The person provides information about the importance of different kinds of activities, satisfaction with current performance, and whether support is wanted. The model includes motives relating to social networks, participation in society, physical health and strength, mental health, performing activities of one's choice, enjoyment, and safety. These categories provide a way of connecting activities to several personally relevant purposes [Paper IV].

The resulting representation makes it possible to distinguish between a **general health objective** and the person's own reasons for an activity. For example, physical activity may contribute to physical health while also supporting social contact, enjoyment, independence, or another personally important motive. These motives need not always point towards the same action. Person-tailored support therefore requires mechanisms for representing priorities and, where relevant, reasoning about competing considerations.

The thesis does not claim that this constitutes a comprehensive model of a person. The representations are task-oriented: they capture selected information needed for the forms of health-related dialogue, coaching, reflection, and planning investigated in the research. Information may also have different

origins, including domain knowledge, observations, baseline assessments, and information provided through dialogue.

The later storytelling work operationalises this modelling direction in a structured user model containing activities, motivations, goals, preferences, social context, and ratings such as importance and fun. Activities are represented across physical, social, and recovery domains. The user model is then used as an input to narrative planning and as the external reference for separately assessing persona grounding [Paper V]. This is a later implementation of the broader thesis principle that generated person-tailored content should remain connected to explicit information about the person rather than relying only on unrestricted generation.

Across these models, person-tailoring is therefore based on relationships among several types of information:

- **activities** describe what the person does or intends to do;
- **goals** describe intended states or outcomes;
- **motives** represent why activities or outcomes matter;
- **preferences and priorities** help distinguish what is important to the person;
- **context** provides information needed to interpret the current situation; and
- **dialogue information** can update or clarify the model during interaction.

These elements form the representational basis on which different dialogue purposes and agent roles can operate.

3.4 Representing Dialogue Purposes and Agent Roles

Representing the person's situation does not by itself determine what an agent should do. The agent also requires a **purpose for the interaction**. A dialogue concerning the same activity can be conducted for different reasons. The agent may need to obtain missing information, investigate a situation, deliberate about possible actions, or provide reasons intended to persuade the person.

Paper I addresses this distinction by combining a topic with a dialogue goal. For example, when a possible problem concerning sleep is selected as a topic, the agent may conduct an information-seeking dialogue to obtain missing information about related factors. The purpose of the interaction determines the questions and dialogue behaviour required rather than the topic alone determining the sequence [Paper I].

The early architecture uses dialogue **schemes** as templates for goal-oriented dialogue tasks. These schemes include mechanisms for dialogue initiation, creation of the belief base, execution of the dialogue body, help, and handling time-outs. They are combined with state transitions to determine dialogue behaviour. Importantly, these dialogue-management schemes are different from the **argumentation schemes** used in the later work. Paper I explicitly distinguishes the higher-level dialogue-strategy schemes from argumentation schemes representing patterns of reasoning [Paper I].

The architecture initially demonstrates information-seeking dialogue, while identifying other dialogue purposes, including inquiry, deliberation, and persuasion, as directions for extending the approach. This distinction is important when interpreting Paper I: it establishes an architectural basis for multiple purposeful dialogues but does not implement and evaluate all of these dialogue types [Paper I].

Related work has also explored multiple virtual coaches with different expertise, using argumentation to present and manage potentially conflicting perspectives while adapting argumentative interaction to the user [47]. Paper IV develops the multipurpose aspect further by connecting motives to **complementary agent roles**. Instead of treating digital coaching as if it had only one objective, the model considers a team of agents with different responsibilities and knowledge domains. Their purposes correspond to different aspects of the person's activities and motives. This provides a computational way of representing situations in which several forms of support may be relevant simultaneously [Paper IV].

The agent roles are not intended to imply that each motive should independently determine the person's behaviour. Rather, they provide a structure for bringing different considerations into reasoning. When motives support different actions or priorities, argumentation can be used to construct reasons that take more than one motive into account. Paper IV develops strategies for selecting supporting arguments and constructing multipurpose arguments from information in the motivational model. The NejaVis prototype provides an initial implementation in which baseline information can be used and extended during argumentation-based dialogue [Paper IV].

This relationship among **activity, motive, dialogue purpose, and agent role** is a central part of the thesis-level model. Activity information identifies what is being considered. Motives and goals represent why it may matter. Dialogue purpose identifies what the interaction is trying to accomplish. Agent roles organise the knowledge and responsibilities relevant to different aspects of the person's situation.

The later storytelling study uses a related separation between **content** and **communicative purpose**. Its narrative planner selects a dialogue type and corresponding argumentation scheme before narrative generation. Persuasion, deliberation, and inquiry are associated with different forms of reasoning rather than being treated simply as different topics [Paper V]. This extends the earlier principle of purpose-directed dialogue into narrative generation, although

Paper V evaluates generated narratives rather than live multi-turn dialogue.

Thus, the models developed across the thesis progressively make the purpose of support more explicit. The purpose is not inferred merely from the fact that an interaction concerns health. It is represented as a distinct element that influences what information is required, which reasoning mechanisms are relevant, and what contribution the agent attempts to make.

3.5 Cognitive Architecture for Person-Tailored Dialogue

The cognitive architecture proposed in Paper I provides an early computational organisation for purposeful human-agent dialogue. It consists of four main components: the **Belief Base, Dialogue Manager, Task Manager, and Plan Generator** as illustrated in Figure 3.1 [Paper I, Paper II]. Each component is assigned responsibilities that contribute to the agent's goal-directed behaviour [Paper I].

The **Belief Base** contains facts and rules available to the agent. The architecture uses semantic models in ACKTUS to provide information relevant to the selected topic. When a dialogue begins, user and domain information can be used to populate the belief base. Missing information can identify questions that the agent may ask. Beliefs can also be prioritised using information from domain experts, priorities and goals stated by the person, and unresolved information from previous dialogue [Paper I].

The **Dialogue Manager** is responsible for determining dialogue behaviour. It uses a state-machine approach in which transitions depend on the dialogue state and the person's responses. Dialogue schemes provide templates that are instantiated within states. The dialogue strategy therefore combines schemes with state transitions rather than requiring one fixed linear dialogue protocol [Paper I].

The proposed **Task Manager** contains situation-specific domain knowledge and reasoning concerning possible next moves. In the example of information-seeking dialogue, it organises questions associated with missing information. The **Plan Generator** is intended to organise plans for achieving dialogue-related goals. Together, these components describe how knowledge, dialogue state, task reasoning, and planning could be separated within a purposeful agent architecture.

However, the implementation status of these components must be distinguished from the conceptual architecture. Paper I presents an **initial prototype of the Dialogue Manager and dialogue strategy**, including schemes and state transitions. The complete cognitive architecture was not implemented. In particular, implementation of the Task Manager and Plan Generator was identified as future work [Paper I]. The contribution should therefore be understood as a proposed architecture with partial implementation rather than a completed adaptive dialogue system.

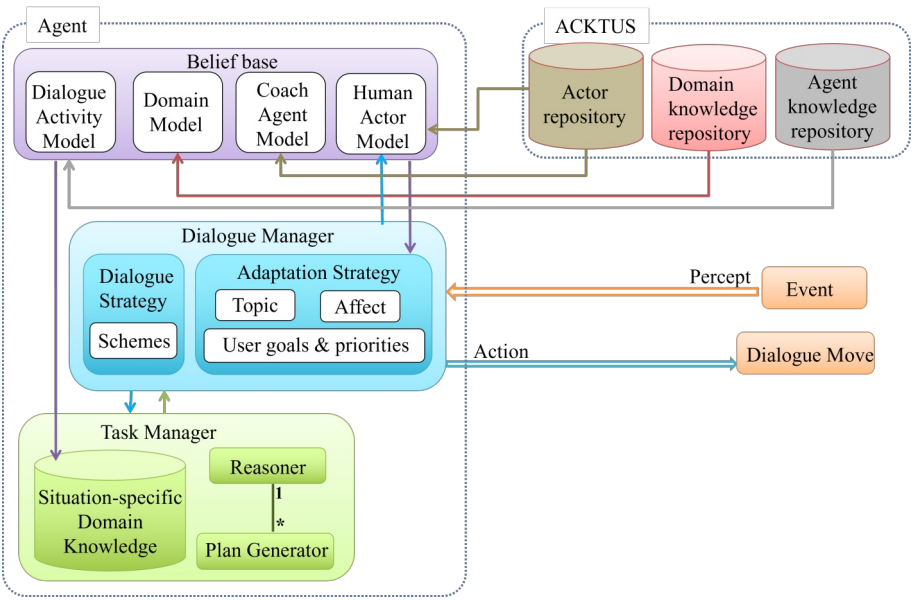


Figure 3.1: Cognitive architecture for a person-tailored software agent supporting human-agent dialogue. The architecture was proposed in Paper I and further developed in Paper II.

The later work does not simply implement all remaining components of Paper I. Instead, different aspects of the architectural problem are developed through subsequent models and prototypes. Paper III strengthens the representation of purposeful activity. Paper IV develops motivational models, complementary agent roles, and argument-based mechanisms for handling multiple purposes. These developments provide richer information and reasoning structures that could support person-tailored interaction without constituting a single completed implementation of the original architecture.

The storytelling architecture in Paper V represents a later technical configuration (refer Figure 3.2). It contains layers for user interaction, the structured user model, knowledge and argumentation constraints, narrative planning and generation, and reflective analysis. It therefore preserves several concerns visible in the earlier architecture, representation of the person, explicit interaction purpose, reasoning structures, and generated agent contributions, while combining them with generative AI [Paper V].

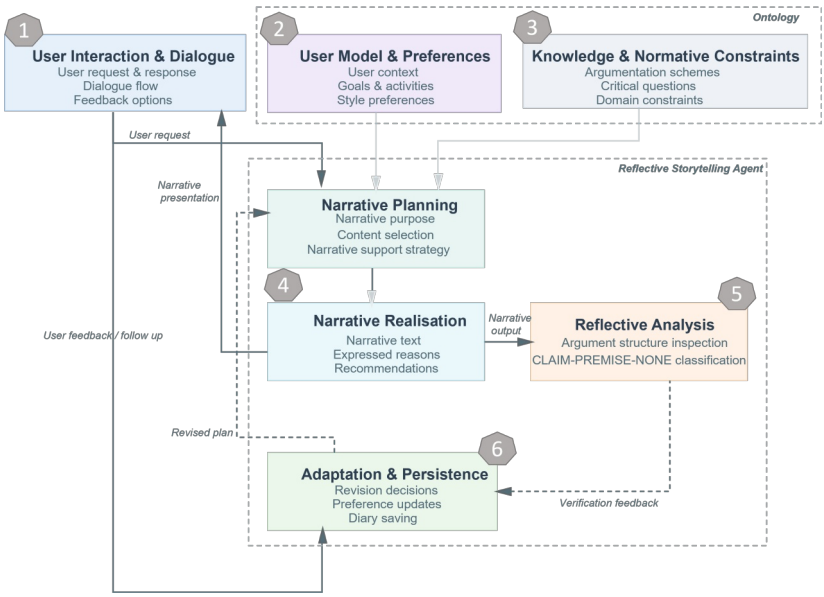


Figure 3.2: System architecture for argumentation-based person-tailored storytelling (Paper V). The architecture integrates activity-based user modelling, argumentation knowledge and reasoning, LLM-based narrative generation, and argument mining for reflective inspection. Dotted arrows indicate prospective feedback and adaptation pathways that were not evaluated in the study.

This later architecture also illustrates an important boundary concerning adaptation. Paper V depicts prospective pathways through which reflective-analysis results and participant feedback could inform future generation (refer Figure 3.2), but these feedback and adaptation pathways were **not evaluated** in that study [Paper V]. For this reason, the thesis uses person-tailored dialogue rather than claiming a fully adaptive architecture whose behaviour has been demonstrated to change autonomously over time.

Across the thesis, cognitive architecture is consequently treated as an organisation of capabilities required for purposeful support rather than as one final implemented system. The research progressively investigates how those capabilities can be represented and realised: maintaining relevant knowledge about a person and situation, identifying interaction purposes, managing dialogue, reasoning with goals and motives, generating person-tailored contributions, and allowing generated content to be inspected.

3.6 Summary

This chapter has described the modelling foundations for person-tailored human-agent dialogue developed across the thesis. The central modelling problem is

to connect **what a person does** with **why the activity matters, what the interaction is intended to achieve**, and **what information and reasoning an agent requires to contribute appropriately**.

Paper III provides a computer-interpretable representation that connects observations and events with purposeful activities, actors, instruments, goals, motives, and context. Paper IV develops an early approach for representing multiple motives and goals and associates these with complementary agent roles and person-tailored argumentation. Paper I provides an architectural organisation for purposeful dialogue through a belief base, dialogue manager, task manager, and plan generator, while only part of this architecture, particularly the dialogue-management mechanisms, was implemented [Paper III, Paper IV, Paper I].

Together, these contributions establish a representational progression from **activity and context**, through **goals and multiple motives**, to **dialogue purpose and agent reasoning**. Later studies build on these foundations by using structured activity and motivation information in argumentation-guided narrative generation and by exploring generated content as an object for reflection and collaboration. The models do not constitute a complete computational representation of a person or a fully implemented integrated dialogue system. Instead, they provide complementary structures for representing and reasoning about selected aspects of purposeful person-tailored support.

Chapter 4 develops the next part of this progression: the mechanisms through which represented purposes, motives, and knowledge are used in **argumentation, dialogue, narrative generation, reflective inspection, and collaborative interaction**.

Chapter 4

Supporting Collaborative Reasoning and Dialogue

The models described in Chapter 3 provide representations of activities, goals, motives, context, dialogue purposes, and agent roles. These representations become useful for person-tailored support when they can guide interaction between a person and an agent. This chapter describes the mechanisms developed in the thesis for this purpose.

The research takes a progressively broader view of human-agent interaction. The early work focuses on structured dialogue in which the person and agent exchange information and reason about health-related topics. The multipurpose model introduces argumentation for reasoning with several motives and goals. Later work uses argumentation to guide LLM-based narrative generation and introduces mechanisms for inspecting generated argumentative structure. The final study explores diary writing as a collaborative activity in which generated content can be interpreted and revised rather than treated as a fixed output.

These developments do not constitute one continuously implemented system. They include conceptual models, partially implemented architectural components, dialogue prototypes, implemented narrative-generation mechanisms, and participatory-design requirements. The common concern is how computational support can remain connected to the person's activities, purposes, and participation.

4.1 Dialogue as Collaborative Activity

Human-agent dialogue in this thesis is not understood only as a sequence of questions and answers. It is a purposeful activity in which a person and a software agent contribute information towards an interaction goal. The dialogue can also change as new information becomes available or as the immediate

purpose of the interaction changes. Recent work on human-LLM agent teams similarly emphasises collaboration through shared goals, coordination, adaptation to human characteristics, transparency, and complementary responsibilities between human and agent [107, 5].

The early dialogue architecture illustrates this through nested dialogue goals. A person may begin with a general concern, while the agent needs additional information about related topics before the broader concern can be addressed. In Paper I, for example, a concern about sleep leads to information-seeking about factors such as pain and medication before returning to the original topic. The dialogue therefore combines a general purpose with specific sub-goals rather than following a single predetermined sequence [Paper I].

Paper II implements and formatively evaluates this interaction approach. The system builds its belief base using information from the user model and relevant domain knowledge. The person and the agent interact through dialogue moves, while information obtained during interaction can modify the user model. The evaluation was restricted mainly to initiation of inquiry dialogues and nested information-seeking dialogues; it therefore provides evidence concerning an early prototype rather than a general implementation of all proposed dialogue purposes [Paper II].

An important feature of collaborative dialogue is **contextual appropriateness**. A contribution may concern a relevant topic but still be inappropriate at the particular point at which it occurs. Paper II therefore asked participants to assess agent moves in the context in which they appeared. The formative study suggested that evaluating dialogue contributions within their immediate context was useful, while also identifying limitations of the evaluation method and the need for further development [Paper II].

Paper III connects this dialogue process more explicitly with purposeful activity. It describes human-agent dialogue itself as an activity that can have a generic goal corresponding to a dialogue type and several sub-goals associated with particular topics. Within a broader interaction, the person and agent can move among sub-dialogues as information is obtained and priorities change [Paper III].

This perspective makes collaboration more than the agent adapting a message to a static profile. The person contributes information, chooses or redirects topics, responds to the agent's interpretation, and can influence the information available for subsequent interaction. At the same time, the degree of collaboration differs substantially across the studies. Paper V, for example, evaluates generated narratives rather than a live collaborative dialogue. It should therefore not be treated as evidence of interactive co-construction.

The collaborative perspective becomes more explicit in Paper VI. Diary writing is considered a collaborative activity in which the person and agent contribute to reflection on previous activities and planning of future activities through the co-creation of diary posts, as illustrated in Figure 4.1. The participatory-design study identifies grounding, explanation, ownership, flexibility, emotional safety, and opportunities to question or revise the agent's

contribution as requirements for this form of interaction.

Figure 4.1 illustrates the collaborative co-creation of diary posts from the perspective of Cultural-Historical Activity Theory, highlighting the roles of the person and agent in this shared activity [33].

Across these studies, collaborative dialogue therefore develops from **exchange and contextual response** towards **shared interpretation and revision**. The agent contributes computational capabilities, but the person's information, goals, judgements, and responses remain part of the process.

4.2 Argumentation Schemes and Dialogue Types

Different dialogue purposes require different forms of reasoning. As described in Chapter 2, dialogue types distinguish interactions according to their communicative goals. The thesis uses this distinction to connect the purpose of an interaction with the reasoning mechanisms available to the agent.

The early architecture focuses primarily on information-seeking within inquiry-oriented dialogue. Subsequent work expands the range of purposes to include inquiry, deliberation, and persuasion. These purposes differ in what the interaction is intended to accomplish. Inquiry concerns examining or establishing information, deliberation concerns considering possible actions, and persuasion concerns providing reasons for adopting or reconsidering a position or action.

Argumentation provides a mechanism for connecting these purposes with explicit reasons. Paper IV develops an early approach to person-tailored argumentation in which arguments are constructed from a motivational model containing several potentially relevant motives and goals. Rather than assuming that one health objective should determine the interaction, the agent can use information about what the person considers important when selecting arguments.

The NejaVis prototype provides an initial implementation of this approach. Baseline information about activities, importance, satisfaction, and desired support forms part of the user model. The dialogue system can use this information when constructing motivational arguments and can update or extend the information during interaction. Paper IV describes nested argumentation-based dialogues in which conclusions may be drawn concerning a condition or possible action, while disagreement can lead the agent to provide arguments related to priorities previously expressed by the person [Paper IV].

The use of **multiple motives** is important here. An activity may contribute simultaneously to physical health, social participation, enjoyment, independence, safety, or other purposes. Argumentation provides a means of making these considerations explicit rather than reducing support to a single objective. Paper IV therefore develops mechanisms for selecting and combining arguments associated with complementary motives. This is a conceptual and prototype contribution that provides a basis for future evaluation of how multipurpose arguments may influence health-related behaviour and outcomes.

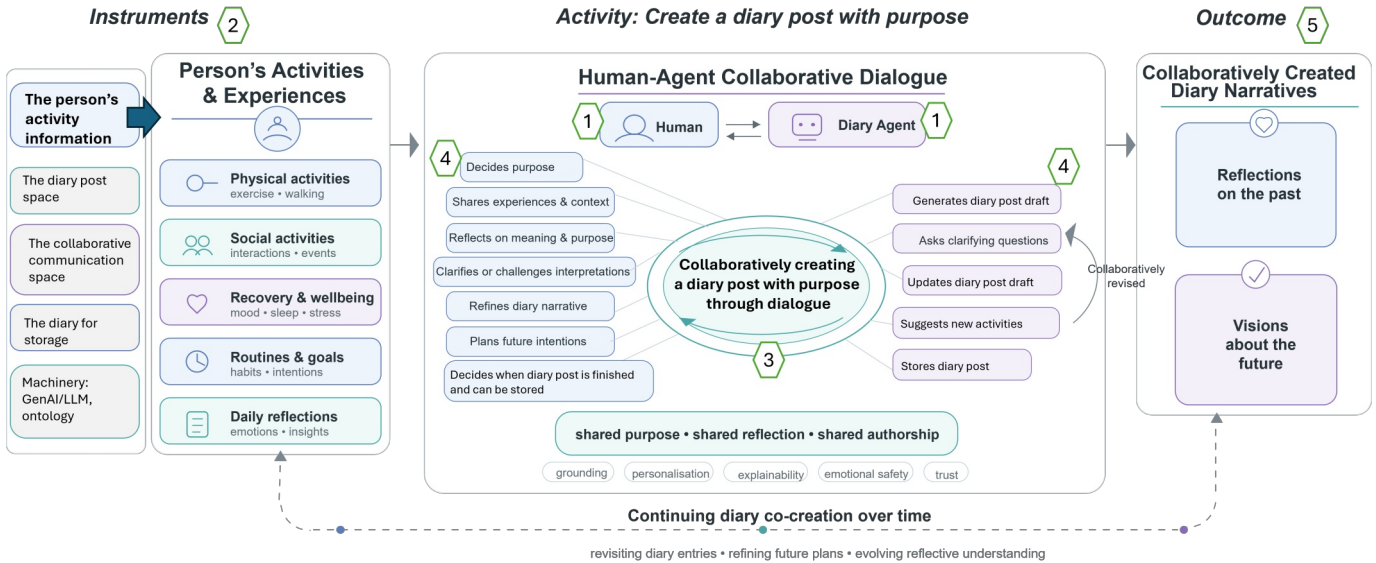


Figure 4.1: Collaborative creation of diary posts with purpose, interpreted through Cultural-Historical Activity Theory [33]. Adapted from Paper VI.

Paper V later uses argumentation schemes in a different technical setting: LLM-based narrative generation. Here, dialogue purposes are mapped to selected reasoning schemes before a narrative is generated. Argument from Value is used for persuasion because it connects a proposed action with personally important values. Sufficient Condition is used for deliberation to structure reasoning about possible actions and outcomes. Position to Know is used for inquiry when reflecting on information available from recorded activities. Argument from Expert Opinion remains represented in the ontology but was not used in the evaluated narratives because the stories did not introduce an external expert source [Paper V].

The argumentation scheme in Paper V therefore acts as **generation guidance**. It provides a reasoning structure that the language model can use when constructing the narrative. The technical evaluation showed that this guidance substantially increased explicit premises while claim counts remained essentially unchanged. However, it did not produce a statistically significant independent improvement in sampled persona grounding [Paper VI].

This distinction is important for the thesis-level contribution. Argumentation schemes can structure **how reasons are expressed**, but their presence does not guarantee that the content of those reasons is supported by information about the person. Argumentative structure and persona grounding are complementary properties requiring separate assessment.

Across the research, argumentation thus serves two related purposes. In the earlier dialogue work, it provides a mechanism for reasoning with a person's goals, priorities, and potentially competing motives. In the later storytelling work, it provides explicit reasoning structures for purpose-guided narrative generation. In both cases, the communicative purpose of the interaction influences the reasoning strategy rather than argumentation being applied independently of dialogue purpose.

4.3 Person-Tailoring, Narrative Generation, and Reflective Inspection

Person-tailoring in the earlier dialogue systems is based mainly on explicit user and domain models. Relevant knowledge is retrieved according to the person and topic, and information obtained during dialogue can update the user model. Paper II, for example, describes a prototype in which the agent constructs its belief base from ACKTUS knowledge repositories according to the topic and the person interacting with the system [Paper II].

Paper IV extends person-tailoring by representing the relative importance of activities, satisfaction with their performance, desired support, and multiple motives. This information is used in prioritisation and in selecting arguments for dialogue. The resulting tailoring therefore concerns not only what topic should be discussed but also which reasons may be relevant to the person's stated priorities [Paper IV]. Storytelling has also been conceptualised as having

purposes beyond entertainment, including applications in health and well-being [67]. Lugmayr et al. describe such "serious storytelling" in terms of purposeful goals, meaningful interaction, and knowledge creation.

The introduction of generative AI in Paper V changes the technical form of this process. Instead of selecting only predefined dialogue contributions or arguments, the system uses an LLM to realise a complete narrative. Structured persona and activity information, dialogue-purpose guidance, grounding instructions, and argumentation schemes are supplied to guide generation. The resulting architecture combines structured knowledge and formal reasoning guidance with the flexibility of language generation [Paper V].

This flexibility also introduces a new problem. A fluent narrative can include person-specific statements that are not supported by the structured information supplied to the system. Consequently, person-tailored generation requires a distinction between **generation guidance, persona grounding,** and **argumentative structure**.

Paper V evaluates these properties separately. Persona grounding is assessed by researchers comparing sampled person-specific statements with the corresponding structured persona information. Argument mining, by contrast, identifies whether textual units function as **CLAIM, PREMISE**, or **NONE**. Argument-mining performance therefore concerns recognition of argumentative function, not whether a statement is factually correct or supported by the persona [Paper V].

For this reason, the mechanism is better described as **reflective inspection** than as agent self-reflection. The argument-mining layer makes aspects of generated argumentative structure available for inspection. It does not demonstrate that the agent understands its own reasoning in a human-like sense, and it does not autonomously verify its output.

The technical results reinforce this limitation. The zero-shot argument miner achieved moderate performance against the human-consensus annotation and had particular difficulty distinguishing non-argumentative content. The latest Paper V therefore treats its predictions as suitable for assisted inspection rather than autonomous verification [Paper V].

Likewise, the grounding analysis should not be described as part of the argument miner. The full structured condition showed substantially greater sampled persona grounding than the standard baseline, while the comparison between the full system and the scheme-ablation condition did not establish an independent grounding improvement from scheme guidance. Unsupported statements remained in the structured conditions [Paper V].

These results motivate a broader view of reflective interaction. Computational inspection can expose aspects of generated structure, but interpretation and correction can also involve the person. Paper VI develops this direction by considering diary entries as shared artefacts that can be discussed and revised. The relevant goal is therefore not to replace human judgement with an autonomous verifier, but to combine structured information, computational inspection, and opportunities for human participation.

This progression connects the two meanings of **reflection** used in the thesis. *Human reflection* concerns a person's interpretation of previous activities, motives, experiences, and possible future actions. *Reflective inspection* concerns computational analysis of generated content. Keeping these meanings separate avoids implying that an AI system has human-like self-reflective capacities.

4.4 From Conceptual Models to Implemented Mechanisms and Prototypes

The six papers differ substantially in their level of technical implementation. Distinguishing these levels is necessary for understanding what the thesis demonstrates.

Paper I proposes a cognitive architecture containing a Belief Base, Dialogue Manager, Task Manager, and Plan Generator. The dialogue-management mechanisms were partially implemented, while the complete architecture was not. Paper II subsequently evaluates a prototype dialogue system containing a belief base, dialogue manager, and task manager and integrated with ACKTUS-based semantic knowledge [Paper I, Paper II].

Paper III shifts attention from dialogue control to knowledge representation. Its core and event ontologies were implemented using RDF/OWL and integrated into prototype applications. The contribution is therefore an implemented representation rather than an evaluated person-tailored dialogue intervention. The paper itself describes further development of proactive person-tailored dialogue behaviour as ongoing or future work [Paper III].

Paper IV brings together a motivational model, complementary agent roles, and argumentation-based dialogue mechanisms. The NejaVis system provides an initial implementation of a Companion Agent that can use baseline information and person-tailored arguments in dialogue. The work demonstrates how the conceptual model can be operationalised, but it does not constitute a validated intervention or establish outcomes for older adults [Paper IV].

Paper V represents a further change in implementation. Its reflective storytelling architecture combines ontology-based user modelling, argumentation schemes, LLM-based narrative generation, and an argument-mining layer. The system generates narratives under explicit communicative purposes and provides mechanisms for inspecting argumentative structure. The study also evaluates grounding separately against structured persona information [Paper V].

Paper V nevertheless remains different from a fully interactive dialogue system. The 55 older adults evaluated previously generated narratives based on fictional personas rather than conducting live collaborative dialogues grounded in their own histories. The study therefore demonstrates and evaluates person-tailored narrative-generation mechanisms under controlled conditions, not a deployed longitudinal companion system [Paper V].

Paper VI moves from completed generated narratives towards collaborative construction of diary posts. It uses participatory design with six occupational-

therapy experts to identify requirements and explore how a person and diary agent might jointly develop reflective content. The resulting prototype is exploratory and only partially implemented. Consequently, requirements such as grounding, explainability, ownership, emotional safety, and flexible revision should be treated as design requirements rather than established system capabilities or effects.

Across the thesis, implementation therefore develops through **different technical artefacts rather than one continuously expanded application**. These include an architectural proposal and partial dialogue manager, a formative dialogue prototype, implemented activity ontologies, a motivational and argumentation-based coaching prototype, an implemented hybrid narrative-generation architecture, and an exploratory collaborative diary prototype.

This distinction is important because the thesis contribution lies partly in how these mechanisms progressively address different aspects of the same research problem. The research moves from controlling purposeful dialogue, through representing activities and multiple motives, towards explicit argumentation, generative narratives, reflective inspection, and collaborative interpretation.

4.5 Summary

This chapter has described how the representational foundations of Chapter 3 are used to support reasoning and interaction. The central development is from **purposeful dialogue**, through **person-tailored argumentation**, towards **argumentation-guided narrative generation, reflective inspection, and collaborative interpretation**.

Dialogue is treated as a purposeful and contextual activity rather than only as an exchange of isolated messages. Early prototypes allow the person and agent to move through nested topics and dialogue goals while information obtained during interaction contributes to the user model. Argumentation extends this approach by providing explicit reasons related to the person's motives, goals, and priorities.

The later storytelling architecture combines structured person information and dialogue purposes with argumentation schemes and LLM-based generation. Argumentation-scheme guidance increases the explicit expression of premises, but it does not independently establish persona grounding. Argument mining provides a mechanism for inspecting argumentative structure and must not be interpreted as factual verification, grounding assessment, or autonomous hallucination detection [Paper V].

Finally, the collaborative diary work broadens the interaction model from receiving generated content towards participating in its interpretation and revision. This introduces requirements concerning human agency, grounding, explanation, ownership, and flexible interaction. Because these requirements

arise from expert participatory design and an exploratory prototype, their effects have not been established with older adults.

Taken together, the work establishes complementary mechanisms for connecting activities and context → motives and goals → dialogue purposes → argumentation → generated narratives → inspection and collaborative revision. The empirical and technical evidence used to evaluate these mechanisms is presented in Chapter 5.

Chapter 5

Empirical Studies and Evaluation

The six papers use complementary methods to address different aspects of the research questions. This chapter describes the methodological approaches and their roles in the overall research.

Papers I, III, and IV primarily contribute conceptual and computational development. Paper I proposes a cognitive architecture for purposeful human-agent dialogue and provides a partial implementation of its dialogue-management components. Paper III develops and implements ontology-based representations for purposeful human activity, connecting observations and events with actors, instruments, goals, motives, and contextual information. Paper IV develops an early approach to modelling multiple motives and goals for personalised digital coaching. It also describes complementary agent roles and mechanisms for generating person-tailored arguments, representing a novel direction at the time of publication in addressing multiple purposes and potentially conflicting motives within personalised digital coaching.

These papers establish architectural and representational foundations, but they do not provide evidence of clinical effectiveness or long-term outcomes.

Papers II, V, and VI provide the main human-centred empirical evidence, although they use different participants and research methods. Paper II reports a formative evaluation of a prototype health-related dialogue system with therapists and older adults. Paper V combines formative evaluation with domain experts, an evaluation of generated narratives with older adults, and a separate technical evaluation of narrative grounding and argumentative structure. Paper VI uses participatory design with occupational-therapy experts to investigate requirements for collaborative diary interaction and develops an exploratory prototype. The findings from these studies are therefore interpreted according to the particular system, participants, and evaluation method used in each study rather than as evidence from a single cumulative trial. This dis-

tinction is important because the thesis combines architectural proposals and implemented mechanisms with formative and exploratory empirical studies.

5.1 Participatory Design and Evaluation engaging Domain Experts

Domain experts contributed to three different empirical components of the thesis. Their roles differed between the studies and the participant groups are therefore treated separately.

Paper II involved five therapists in the formative evaluation of an early human-agent dialogue prototype. The evaluation examined dialogue purposes, topics, interaction design, and the appropriateness of individual agent moves. Data were obtained through methods including observation, interviews, a therapist focus-group discussion, interaction logs, and ratings of agent moves during dialogue. A particular concern was whether an agent contribution was appropriate not only to the general topic but also to the immediate dialogue context. The study provided formative evidence for subsequent design.

Paper V involved eleven domain experts in Phase I of the storytelling study. The participants were health professionals with experience related to older adults and/or health promotion. They included occupational-therapy, physiotherapy, and nursing expertise. During individual formative co-design sessions, experts provided activities, evaluated generated narratives covering different dialogue purposes and creativity conditions, and discussed aspects including communicative purpose, creativity, motivational value, relevance, potential risks, and improvements. Their feedback informed iterative refinement of the storytelling system and study protocol before the evaluation with older adults [11].

In Paper VI participatory design was conducted with six occupational-therapy experts to investigate human-agent collaboration in diary writing. The study combined co-design activities with interviews, observation of prototype use, and researcher notes. The resulting material was analysed to identify requirements and design implications for collaborative reflection and planning. The study identified requirements concerning grounding, personalisation, explainability, emotional safety, ownership, flexibility, and meaningful participation in the construction of diary content. These findings describe expert-informed requirements and directions for design. They do not establish that all of these capabilities were implemented successfully or that they produce beneficial outcomes for older adults.

Taken together, these studies engaging experts served different methodological purposes. Paper II evaluated an early dialogue prototype. In Paper V experts were engaged to refine person-tailored storytelling before the older-adult study, and in Paper VI experts were engaged in participatory exploration of collaborative diary interaction. The results should consequently be interpreted as formative and design-oriented rather than as clinical validation.

5.2 Evaluation with Older Adults

Older adults participated in the evaluation studies reported in Papers II and V, examining different forms of interaction.

The study in Paper II included eleven older adults in its formative evaluation of health-related human-agent dialogue. Participants interacted with a restricted implementation of the dialogue system and contributed assessments of dialogue topics, purposes, interaction design, and the appropriateness of individual agent moves. The findings distinguished topic-related appropriateness from contextual appropriateness. An agent move could concern a relevant health topic while still being inappropriate in the particular point of the dialogue. The study therefore provided early empirical support for treating person-tailored dialogue as a situated interaction rather than only as the retrieval of information related to a user profile or topic. The small sample and restricted implementation limit the conclusions that can be drawn from this study.

Paper V substantially extended the older-adult evaluation study in Paper II. Phase II involved 55 adults aged 66–83 from 10 European countries, recruited through the Prolific platform. The participants evaluated narratives associated with inquiry, deliberation, and persuasion purposes and different creativity conditions. Measures included perceived narrative purpose, creativity and appreciation, perceived inconsistencies, relevance, cultural relatability, and intention to use similar functionality in the future [11].

An important methodological limitation is that these narratives were **not generated from the participants' own personal histories**. Participants selected the most similar of five predefined fictional personas. The personas contained structured activity and motivation profiles, and the narratives were generated from this predefined information. The study therefore examines older adults' perceptions of persona-based person-tailored narratives. It does not demonstrate how the same participants would respond to narratives generated from their own longitudinal personal data [11].

The relationship between computationally specified dialogue purpose and perceived purpose was mixed. After the two inquiry conditions were merged, the analysis contained 330 narrative-purpose assessments. The intended purpose was recognised in 101 assessments (30.6%). Another 110 assessments (33.3%) indicated no clear purpose, while 119 (36.1%) attributed another purpose. Multiple purpose selections were possible. These results show that specifying a communicative purpose during generation does not ensure that readers will interpret the narrative according to that purpose.

The evaluation also produced mixed evidence concerning acceptance and future use. 24 of the 55 participants described the narratives as culturally relatable, seventeen answered "Maybe", and fourteen answered "No". Nine participants indicated that they would use similar functionality in the future, sixteen answered "Maybe", and thirty answered "No". These findings are descriptive and should not be interpreted as evidence that the storytelling ap-

proach improves health, behaviour, or well-being.

5.3 Technical Evaluation of Argumentation-Based Narratives

Paper V complements the human-centred studies with a separate technical evaluation. This evaluation examines two properties that must be distinguished: whether generated person-specific statements are grounded in the structured persona information, and how argumentative functions are expressed and detected in the narratives. Argumentative structure and persona grounding are related concerns, but one does not establish the other [11].

Three matched generation conditions were compared. Condition A, the full system, included structured persona and activity information, grounding instructions, dialogue-purpose guidance, and argumentation-scheme guidance. Condition B was a scheme-ablation condition containing the same persona information, grounding instructions, and dialogue-purpose guidance but without argumentation-scheme guidance. Condition C was a standard LLM baseline supplied with the matched topic, style, and format but without the structured persona information, grounding instructions, dialogue-purpose guidance, or argumentation schemes. The A–B comparison therefore examines the incremental contribution of scheme guidance, whereas the A–C comparison concerns the structured architecture as a package and cannot isolate the effect of any one component [11].

Persona grounding was assessed separately from argument mining. A sample of 120 statements, with 40 from each generation condition, was assessed by three researchers against the corresponding predefined persona information. Statements were classified according to whether person-specific content was supported by the persona information or introduced content that was unsupported or otherwise not assessable. Final reference labels were established through majority judgement, with unresolved cases discussed by the annotators [11].

Among applicable sampled statements, 67.6% in the full-system condition were grounded in persona information compared with 57.9% in the scheme-ablation condition. This difference was not statistically significant. The results therefore do not establish an independent improvement in persona grounding from argumentation-scheme guidance. The full structured condition showed substantially greater sampled grounding than the standard baseline, but because several components differ simultaneously between these conditions, this comparison supports the structured architecture as a package rather than attributing the difference specifically to argumentation schemes.

Argumentation schemes had a clearer effect on explicit argumentative organisation. The full-system narratives contained substantially more explicit premises than the matched scheme-ablation narratives, while the number of claims remained essentially unchanged. This provides evidence that scheme

guidance affected how reasons were expressed in the narratives, rather than evidence that it independently improved persona grounding.

A separate evaluation examined whether a zero-shot argument miner could identify CLAIM, PREMISE, and NONE units in generated narratives. Human-consensus annotations were created for the full-system and scheme-ablation datasets. The resulting gold standard contained 529 argumentative discourse units for the full-system condition and 436 for the scheme-ablation condition. The zero-shot model was evaluated against these annotations using measures including precision, recall, F1-score, accuracy, and confusion matrices [11].

The argument miner achieved moderate rather than sufficient performance for autonomous assessment. Its role in the architecture should therefore be understood as a means of inspecting argumentative structure. It does not determine whether a statement is factually correct, verify whether it is grounded in persona information, detect hallucinations autonomously, or establish general argument quality. Persona grounding requires a separate comparison with external structured information. This separation is central to interpreting the technical evaluation correctly [11].

5.4 Cross-Study Findings

Across the studies, a recurring finding is that person-tailoring depends on more than selecting information that is broadly related to a person or topic. Paper II shows this at the level of individual dialogue moves: topic-related relevance does not necessarily imply appropriateness in the immediate dialogue context. Paper V shows a related distinction at the narrative level: a formally specified dialogue purpose does not determine the purpose perceived by a reader. These studies use different systems and evaluation methods, but together they demonstrate a recurring difference between properties represented or intended by the system and how its contributions are interpreted by people.

The conceptual and technical studies provide mechanisms for representing more of the context required for such interaction. Paper III connects activities with actors, instruments, observations, goals, motives, and contextual evidence. Paper IV extends this direction by representing multiple motives and priorities and by using them to construct person-tailored arguments. These contributions provide a basis for representing not only what a person does but also why an activity may matter and how several motives may support or conflict with one another. Paper IV remains a model and prototype contribution rather than evidence that multipurpose reasoning produces improved outcomes in practice.

Paper V brings these strands into generative storytelling. Its findings show that structured person information, dialogue-purpose guidance, and argumentation can be incorporated into LLM-based narrative generation, but the resulting properties need separate evaluation. Argumentation-scheme guidance clearly increased the expression of explicit premises but did not produce a statistically significant independent improvement in sampled persona grounding.

Similarly, specifying a narrative purpose did not ensure that older adults perceived the intended purpose. These findings caution against treating formal or computational properties as direct substitutes for human evaluation.

Paper VI further develops the role of human participation. Rather than treating a generated narrative only as a completed output to be evaluated, the study conceptualises diary writing as a collaborative activity in which the person can contribute to interpretation and revision. The occupational-therapy experts emphasised requirements such as grounding, explanation, ownership, emotional safety, flexibility, and the possibility of questioning or correcting the agent. These are design requirements derived from participatory work, not demonstrated effects of a completed system.

Across the thesis, the evidence therefore points towards a progression from representing purposeful activity, to reasoning about multiple motives and dialogue purposes, to generating and inspecting person-tailored narratives, and finally to supporting collaborative interpretation and revision. This progression describes the relationship among the studies; the broader implications are discussed in Chapter 7.

5.5 Summary

This chapter has described the complementary methods and evidence used across the thesis. Papers I, III, and IV primarily provide conceptual, representational, architectural, and computational development. Paper II provides formative evidence from therapists and older adults concerning health-related dialogue and contextual appropriateness. Paper V combines formative expert evaluation, an older-adult evaluation of persona-based narratives, and controlled technical assessment of persona grounding and argumentative structure. Paper VI contributes participatory-design evidence and requirements for collaborative diary interaction.

Together, these methods support conclusions about the design of person-tailored human-agent interaction, the representation of purposeful activities and multiple motives, the importance of dialogue context and perceived purpose, the effects and limits of argumentation-guided narrative generation, and requirements for collaborative interaction. They also expose important limitations. The studies do not constitute a clinical trial of one integrated system, do not establish long-term behavioural or health outcomes, and do not demonstrate a fully implemented and comprehensively evaluated human-agent system. These boundaries define the level at which the thesis findings and contributions should be interpreted.

Chapter 6

Summary of Contributions

The six papers included in this thesis are presented below in an order that reflects the development of the research. Together, they develop and examine complementary aspects of person-tailored human-agent dialogues and narratives to support health and well-being in older adults. Papers I and II establish early architectural and empirical foundations for health-related human-agent dialogue. Papers III and IV provide complementary conceptual and computational bases for the later work: Paper III concerns the representation of purposeful activity and context, whereas Paper IV concerns argumentation-based reasoning about multiple motives, goals, and priorities. Paper V integrates these strands in person-tailored storytelling, and Paper VI extends the approach towards collaborative construction of diary narratives. Table 6.1 provides an overview of the included papers and their roles in the thesis.

Table 6.1: Overview of the included papers and their roles in the thesis.

Paper	Main focus	Main contribution	Role in the thesis
Paper I	Cognitive architecture for human-agent dialogue	Defines a cognitive architecture comprising a belief base, dialogue manager, task manager, and plan generator, together with dialogue strategy schemes.	Provides the initial architectural basis for purposeful and context-sensitive human-agent dialogue.
Paper II	Formative evaluation of health-related human-agent dialogues	Reports older adults' and therapists' experiences of a dialogue prototype, including dialogue purposes, interaction design, and appropriateness.	Shows how health-related dialogue support is experienced by intended users and domain experts.
Paper III	Representation of purposeful human activity	Introduces an instrument-oriented approach and supplementary ontology models for representing activities, events, instruments, observations, evidence, goals, and motives.	Provides a representational basis for grounding dialogue in a person's purposeful activities and their context.

Paper	Main focus	Main contribution	Role in the thesis
Paper IV	Multiple motives, goals, and arguments in digital coaching	Develops a multipurpose motivational and goal model for agent teams, including person-tailored arguments related to potentially conflicting motives.	Provides the reasoning and prioritisation basis for person-tailored support that considers multiple motives, priorities, and goals.
Paper V	Argumentation-based personalised storytelling for older adults	Presents a hybrid storytelling framework combining ontology-based user modelling, argumentation schemes, LLM-based narrative generation, and argument mining.	Integrates structured user modelling and argumentation in a person-tailored storytelling application and examines the relationship between human experiences and computational indicators of narrative grounding and argumentative structure.
Paper VI	Collaborative diary interaction	Examines how a diary agent can support reflection and planning through collaborative construction of narratives grounded in a person's activities, experiences, and motives.	Extends Paper V towards more active human-agent collaboration and identifies design requirements concerning grounding, explainability, ownership, emotional safety, and adaptive interaction.

6.1 Paper I

> Jayalakshmi Baskar and Helena Lindgren. (2014). **Cognitive Architecture of an Agent for Human-Agent Dialogues.** In: Corchado, J.M., et al. Highlights of Practical Applications of Heterogeneous Multi-Agent Systems. The PAAMS Collection. PAAMS 2014. Communications in Computer and Information Science, vol 430. Springer, Cham. `https://doi.org/10.1007/978-3-319-07767-3_9`

Paper I [7] addresses how an intelligent agent can conduct purposeful dialogues with a person about health-related topics. Such dialogues may involve several connected topics, changing user priorities, incomplete information, and different purposes. The paper therefore moves beyond a fixed question-answer dialogue model. It proposes an architecture that enables an agent to organise and adapt its dialogue behaviour according to the evolving situation.

The main contribution is a cognitive architecture for a Coach Agent. The architecture consists of four components: a belief base, a dialogue manager,

a task manager, and a plan generator. The belief base combines information about the person, relevant domain knowledge, and information from previous dialogues. The dialogue manager selects and adapts dialogue moves. The task manager represents situation-specific possibilities and supports reasoning about the purpose of a move. The plan generator creates and revises dialogue plans as the dialogue develops. Together, these components provide a basis for an agent that can handle dialogues that are purposeful, topic-driven, and context-dependent.

The paper also introduces dialogue strategy schemes for managing different dialogue purposes, including inquiry, information seeking, deliberation, and persuasion. In the proposed approach, a dialogue can involve several of these purposes and move between related topics. This extended the focus beyond formal dialogue systems that typically modelled individual dialogue types to investigate their formal properties [16]. The proposed architecture was illustrated through a partial prototype implementation using ACKTUS knowledge repositories [65] and a Java-based agent framework. At this stage, the dialogue manager was only partly implemented, while the task manager and plan generator were identified as further work.

Within this thesis, Paper I provides the architectural starting point for person-tailored human-agent dialogue. It establishes the idea that tailored support requires an agent to combine information about the person with domain knowledge, identify the purpose of a dialogue, and adapt its next move as the situation develops. The later papers extend this foundation by examining how such dialogues are experienced, how activities and motives can be represented, and how person-specific information can support collaborative reflection and personalised narratives.

6.2 Paper II

> Jayalakshmi Baskar and Helena Lindgren. (2015). **Human-Agent Dialogues on Health Topics - An Evaluation Study**. In: Bajo, J., et al. Highlights of Practical Applications of Agents, Multi-Agent Systems, and Sustainability - The PAAMS Collection. PAAMS 2015. Communications in Computer and Information Science, vol 524. Springer, Cham. `https://doi.org/10.1007/978-3-319-19033-4_3`

Paper II [8] examines how older adults and therapists perceive a prototype system for human-agent dialogues about health-related issues in a home environment. Building on the architectural direction of Paper I, the study moves from the design of purposeful dialogue to an initial examination of its use and perceived relevance. The prototype was based on semantic knowledge repositories and a user model containing information about the person's needs, preferences, goals, and priorities. It supported the initiation of *inquiry* dialogues and nested *information-seeking* dialogues, in terms of Walton and Krabbe [98].

The paper reports a pilot evaluation with five therapists and 11 older adults. Data were collected through observation, interviews, focus-group discussion with therapists, interaction logs, and participants' ratings of individual agent moves during the dialogue. The study examined the perceived purposes and topics of dialogue, interaction design issues, and the appropriateness of the agent's behaviour. The small sample and limited subset of implemented dialogue models mean that the findings are indicative and intended to guide further development.

A central contribution of the paper is its method for collecting users' assessments of the appropriateness of each agent move in the context in which it occurred. The analysis distinguished between *context-related appropriateness*, concerning whether a move fitted the immediate dialogue situation, and *topic-related appropriateness*, concerning whether the move was relevant to the overall dialogue topic. The results showed that some inappropriate moves arose when the agent did not adequately interpret a participant's response or update its understanding of the dialogue. This highlighted the need for dialogue strategies that respond to the unfolding interaction, rather than merely following a predefined sequence.

Within this thesis, Paper II provides the first empirical examination of the early dialogue approach. It shows that person-tailored dialogue is not only a matter of storing information about a person or selecting a relevant topic. The agent must also make contextually appropriate moves, present information in an accessible way, and support dialogue topics that users and professionals consider meaningful. These insights motivate the later work on representing everyday activity, reasoning about motives and priorities, and developing more collaborative forms of person-tailored interaction.

6.3 Paper III

> Jayalakshmi Baskar, Chunli Yan, and Helena Lindgren. (2017). **Instrument -Oriented Approach to Detecting and Representing Human Activity for Supporting Executive Functions and Learning**. In Proceedings of the European Conference on Cognitive Ergonomics (ECCE '17) (pp. 105-112). ACM. `https://doi.org/10.1145/3121283.3121305`

Paper III [9] addresses how an intelligent support system can represent and interpret human activity in a way that is meaningful for person-tailored assistance. The paper considers how observations of activity can be connected to the person, the purpose of the activity, and the context in which it takes place. It therefore adopts an activity-centred perspective, informed primarily by Activity Theory [48, 96] and supported by the International Classification of Functioning, Disability and Health (ICF) [1].

[1] https://www.who.int/standards/classifications/international-classification-of-functioning-disability-and-health

The main contribution is an instrument-oriented model for representing purposeful activity and relating observations to higher-level activity interpretations. The model distinguishes three categories of activity according to the instruments that mediate it: activities primarily constrained by the *physical or physical-virtual* environment, by *domain* knowledge, or by *social* knowledge. It represents activity-related events, actors, observations, evidence, goals, motives, and the instruments involved. The model was implemented as two supplementary ontologies: a generic event ontology and a core ontology that can be extended into domain ontologies. This distinction has implications for the design of person-tailored human-agent interaction, since the same observable activity may need to be interpreted differently depending on the actor, goals, motives, and context represented by the system.

The paper illustrates how low-level observations, such as sensor data or interactions with physical and digital objects, can be associated with higher-level activities. It also shows how an assessment protocol can function as an instrument in a domain-specific activity, and how human-agent dialogue can be understood as a socially mediated activity with a generic goal and topic-specific sub-dialogues. These examples demonstrate that the same activity may be conducted in different ways and that support needs to account for this variation.

Within this thesis, Paper III provides a conceptual and technical basis for grounding person-tailored interaction in a person's everyday activities and context. This distinction between physical, knowledge-based, and social aspects is also relevant to the design of human-AI interaction, where support may need to account for combinations of these aspects when interpreting a person's activity and context. It does not evaluate the model with users or demonstrate that it improves support. Rather, it provides the representational basis needed for an agent to connect observations and interaction histories with purposeful activity, goals, and motives. This basis is taken forward in the later work on motive-sensitive reasoning, collaborative diary interaction, and person-tailored narrative generation.

6.4 Paper IV

> Jayalakshmi Baskar, Rebecka Janols, Esteban Guerrero, Juan Carlos Nieves, and Helena Lindgren. (2017). **A Multipurpose Goal Model for Personalised Digital Coaching**. In: Montagna, S., Abreu, P., Giroux, S., Schumacher, M. (eds) Agents and Multi-Agent Systems for Health Care. A2HC AHEALTH 2017 2017. Lecture Notes in Computer Science(), vol 10685. Springer, Cham. `https://doi.org/10.1007/978-3-319-70887-4_6`

Paper IV [10] addresses a central challenge in person-tailored digital coaching: a person may pursue several motives and goals at the same time, and these may conflict. For example, an activity that supports physical health may com-

pete with preferences related to social participation, enjoyment, independence, or feeling safe. The paper therefore questions coaching approaches that focus on a single behavioural objective and explores how support can take account of multiple motives behind a person's activities.

The main contribution is a multipurpose motivational model that is translated into a multipurpose goal model for a team of assistive agents. At baseline, the person indicates the importance of different activities, their satisfaction with current performance, and whether they want support. This information is used to construct a user model and to prioritise relevant motives and goals in a particular situation. The model includes motives related to social participation, participation in society, physical and mental health, self-chosen activities, enjoyment, and safety.

The paper further outlines a team of agents with complementary roles and knowledge sources. A Companion Agent maintains an overall view of the person's situation and coordinates support from the other agents. It uses the multipurpose goal model to select and combine encouraging arguments that relate to the person's priorities and current motivation. The approach was implemented in a prototype coaching system and illustrated through a use case from an earlier pilot study. It was not evaluated as a completed intervention with older adults; formalisation of the model and longer-term user studies were identified as future work.

Within this thesis, Paper IV provides an argumentation-based reasoning and prioritisation basis for person-tailored dialogue. Published in 2017, its multipurpose goal model and team of agents with complementary roles represented, to our knowledge at the time, an early approach to supporting several potentially conflicting health-related purposes within one person-tailored agent system. The paper develops the argument that tailoring should not only mean responding to a person's stored preferences or activity data. It should also consider the person's multiple motives, their relative importance, and possible tensions between them. This perspective is carried forward in Papers V and VI, where motives and priorities inform narrative generation, reflection and planning.

6.5 Paper V

Jayalakshmi Baskar, Vera C. Kaelin, Kaan Kilic, and Helena Lindgren. (2026) **Argumentation -Based Person-Tailored Storytelling for Older Adults: A Reflective Hybrid AI Architecture**. arXiv:2605.10531 [cs.AI]. `https://doi.org/10.48550/arXiv.2605.10531`

Paper V [11] investigates how structured person information, argumentation, and large language models can be combined to generate purposeful person-tailored narratives for older adults. The paper presents a reflective hybrid AI architecture that integrates knowledge-graph-based user modelling, dialogue

purposes, argumentation schemes, LLM-based narrative generation, and argument mining. Structured information about activities and motivations is used to guide person-tailored generation, while argumentation schemes provide reasoning structures associated with different communicative purposes. Argument mining is used after generation to inspect argumentative structure. Persona grounding is assessed separately against the structured persona information.

The study consists of three phases. Phase I used a participatory design methodology with 11 domain experts in a formative evaluation that informed iterative refinement of the storytelling system and study protocol. Phase II involved 55 older adults aged 66–83 who evaluated narratives generated for selected fictional personas; participants' personal information was not collected for narrative generation in this study. The narratives represented inquiry, deliberation, and persuasion purposes at two creativity levels. Participants evaluated aspects including perceived purpose, creativity, relevance, cultural relatability, perceived inconsistencies, and intention to use similar functionality in the future.

The Phase II findings show that the communicative purpose specified during generation was not necessarily perceived as intended. After merging the two inquiry conditions, the analysis contained 330 narrative-purpose assessments. The intended dialogue purpose was recognised in 30.6% of these assessments, while 33.3% were categorised as having no clear purpose and 36.1% as expressing another purpose. The evaluation also showed variation in perceived relevance, cultural relatability, and future-use intention, illustrating that technically specified personalisation and purpose do not by themselves determine how a generated narrative is experienced by a person.

Phase III provided a separate technical evaluation using three matched generation conditions. Condition A, the full system, included structured persona information, grounding instructions, dialogue-purpose guidance, and argumentation -scheme guidance. Condition B retained the same components except for argumentation -scheme guidance. Condition C was a standard LLM baseline without the structured persona information, grounding instructions, dialogue-purpose guidance, or argumentation schemes. The comparison between Conditions A and B therefore examined the incremental effect of argumentation-scheme guidance on the argumentative structure of the generated narratives, particularly the inclusion of explicit premises, whereas the comparison between Conditions A and C examined the structured person-tailored architecture as a package against standard generation.

Persona grounding and argumentative structure were evaluated separately. In a researcher assessment of sampled statements, 67.6% of applicable statements from the full-system condition were grounded in the corresponding persona information, compared with 57.9% in the scheme-ablation condition. This difference was not statistically significant and therefore does not establish an independent improvement in persona grounding from argumentation-scheme guidance. As expected, the full structured condition showed substantially greater sampled persona grounding than the standard baseline, since the base-

line was not provided with persona-related information. This comparison therefore reflects the combined structured architecture rather than the isolated effect of argumentation schemes.

Grounding relies on person-specific information rather than on argumentation-scheme guidance. Argumentation-scheme guidance had a clearer effect on argumentative organisation. Narratives generated with scheme guidance contained on average 6.13 explicit premises per narrative, compared with 1.23 without scheme guidance, while claim counts remained essentially unchanged.

A separate argument-mining evaluation used a human-consensus gold standard in which argumentative discourse units were labelled as CLAIM, PREMISE, or NONE. A zero-shot argument miner was evaluated against these annotations and showed moderate performance. The argument miner is therefore treated as an inspection aid for argumentative structure rather than an autonomous verifier. In particular, argument mining does not establish factual correctness, assess persona grounding, or function as a hallucination detector in our study.

Within this thesis, Paper V brings together several strands developed in the earlier work. It connects structured representations of purposeful activity and motives with dialogue purposes and argumentation, and extends these mechanisms to LLM-based narrative generation. Its thesis-level contribution is to demonstrate how structured person information and explicit reasoning guidance can be incorporated into person-tailored narrative generation while keeping persona grounding and argumentative structure available for separate evaluation. The findings also motivate the later move towards collaborative interpretation and revision explored in Paper VI.

6.6 Paper VI

> Jayalakshmi Baskar, Vera C. Kaelin, and Helena Lindgren. (2026) **Understanding human-diary agent collaboration through an activity-theoretical framework**. Manuscript under minor revision, Journal of Ambient Intelligence and Humanized Computing.

Paper VI [6] examines how a generative-AI-based diary agent can support reflection and planning through dialogue grounded in a person's activities and motives. It addresses a limitation identified in systems that generate completed narratives with limited user participation. The paper instead explores diary creation as a collaborative process in which the person and agent can construct, review, and refine a diary post together.

The paper contributes an activity-theoretical conceptualisation of human-diary agent collaboration. From this perspective, creating a diary post is a purposeful and socially situated activity. The person and diary agent collaborate through dialogue, while the evolving diary post functions as a shared artefact. The dialogue is therefore the collaborative process, whereas the diary post is the evolving outcome of that process.

The study used participatory design with six occupational therapy experts. Data were collected through co-design sessions, interviews, observations of prototype use, and researcher notes. The data were analysed using inductive content analysis and interpreted through an activity-theoretical framework. The prototype was exploratory, and the study did not evaluate long-term outcomes or include older adults.

The findings identify requirements for collaborative diary interaction. These include grounding the agent's contributions in the person's activities and context, explainability, emotional safety, ownership of diary content, adaptive interaction, and meaningful user participation. The findings also identify tensions between reflection and planning, structure and flexibility, and guidance and autonomy. Experts expected opportunities to challenge the agent's assumptions, negotiate interpretations, and revise the narrative. Argument-based explanations emerged as an important requirement for further development.

Within this thesis, Paper VI extends the person-tailored narrative approach examined in Paper V towards more active human-agent collaboration. It brings together the representation of purposeful activity in Paper III, the attention to motives and priorities in Paper IV, and the narrative-generation approach examined in Paper V. The person is no longer primarily the recipient of a generated narrative but an active participant in constructing and refining a diary post with the agent. Paper VI therefore shifts the thesis emphasis from generating person-tailored content towards supporting shared interpretation, negotiation, ownership, and control.

6.7 Overall Contribution of the Thesis

This thesis brings together six papers that develop and examine an approach to person-tailored human-agent dialogues and narratives for health and well-being in older adults. The research progresses from a cognitive architecture for managing purposeful dialogue, through models for representing a person's activities and reasoning about multiple motives and goals, to argumentation-based LLM storytelling and collaborative diary interaction.

At the thesis level, the work makes three connected contributions. First, it develops architectural, conceptual, and computational foundations for grounding person-tailored dialogue and narrative support in structured representations of a person's activities, context, motives, goals, and preferences. This includes an early multipurpose model in which agents with complementary roles coordinate support for several health-related purposes.

Second, the thesis develops argumentation as a mechanism for connecting person-specific knowledge with the purpose of the agent's support. Argumentation is used to represent and reason about motives, goals, values, and potentially competing priorities; to guide deliberative and persuasive support; and, in the LLM-based system, to structure narrative generation and inspect generated arguments and their grounding. The thesis shows how computational

indicators can complement human judgements by revealing points of convergence and divergence.

Third, the thesis provides empirical and design-based knowledge about how intended users and domain experts perceive and experience person-tailored human-agent support. Across the studies, important conditions include contextual appropriateness, grounding in lived activity, understandable purpose, cultural recognisability, explainability, emotional safety, ownership, and opportunities for meaningful participation. The later work shows that person-tailored support should not end with automatic generation. Collaborative co-construction of narratives is also required as a way for the person to question, negotiate, revise, and retain control over the resulting content.

Together, the papers suggest that person-tailored human-agent support for health and well-being should be grounded in structured representations of everyday activity and motives, guided by explicit dialogue purposes and argumentation, and designed to preserve the person's participation, ownership, and ability to examine the agent's contributions. The collective contribution is a knowledge-based and argumentation-oriented approach that connects representation, reasoning, generation, inspection, and collaborative interaction in the development of person-tailored support for older adults, grounded in theory on purposeful human activity.

Chapter 7

Discussion

This thesis investigates how human-agent interaction can be person-tailored to support health and well-being, with a focus on older adults. Across the included work, person-tailoring is approached as more than adapting content to observed behaviour or profile characteristics. The central thesis-level contribution is an approach grounded in Activity Theory in which everyday activities are understood as purposeful and situated, and in which information about activities is connected to goals, motives, priorities, context, and interaction purposes [34, 49]. These representations are subsequently used in dialogue and argumentation mechanisms, extended to generative narratives, and complemented by mechanisms for inspecting and collaboratively interpreting generated content. The resulting contribution is therefore not a single dialogue system or intervention, but a progression of computational models, mechanisms, prototypes, and empirical studies concerned with how AI support can better account for what activities mean to a particular person.

A central argument emerging across the thesis is that **purpose is important at several interconnected levels**. Human activities are purposeful because they are undertaken in relation to goals and motives. Human-agent interactions also have purposes: agents have goals such as seek information, support inquiry about a situation, deliberate about possible actions, or provide reasons intended to persuade. These two levels meet when the agent uses information about a person's activities and motives to determine what support may be relevant and what reasons can be offered. The multipurpose modelling and argumentation work further shows why this relationship cannot always be reduced to a single health objective: the same activity may serve several motives, and these motives may reinforce or conflict with one another. Paper IV operationalises this idea through a multipurpose motivational model, complementary agent roles, and person-tailored multipurpose arguments, while leaving empirical validation of the model as future work [Paper IV].

Argumentation forms an important link between the representational and interactional parts of the thesis. It provides explicit structures for relating

claims and proposed actions to reasons and for differentiating interaction according to dialogue purpose. In the later work, this role is extended from structured human-agent dialogue to generated narratives. Paper V uses dialogue purposes and argumentation schemes to guide LLM-generated narratives based on structured activity and motivation information. The results refine what can be claimed for this approach: scheme guidance substantially affected the explicit argumentative organisation of the narratives, particularly the presence of premises, but did not independently produce a statistically significant improvement in sampled persona grounding. The study therefore distinguishes **argumentative structure from grounding** rather than treating explicit reasoning as evidence that generated content is correct or person-appropriate [Paper V].

A second thesis-level argument concerns the limits of what can be established from the system side. Representing a person's activities and motives does not establish that the representation fully captures what those activities mean to the person. Selecting a formally appropriate topic or dialogue purpose does not ensure that an interaction will be experienced as appropriate. Likewise, grounding generated statements in structured person information does not establish that a narrative will be perceived as relevant, culturally fitting, or useful. Paper V makes this distinction particularly visible: although narratives were generated according to predefined dialogue purposes, the intended purpose was recognised in only 30.6% of the 330 purpose assessments, while participants identified another purpose or no clear purpose in the remaining assessments [Paper V]. These findings extend an issue already encountered in the earlier formative dialogue work: computational intention and human interpretation are related, but they are not equivalent.

The trajectory of the thesis consequently moves towards a more reflective and collaborative understanding of person-tailoring. Automated mechanisms can structure reasoning and make selected properties of generated output available for inspection, but their role needs to remain bounded. In Paper V, argument mining is used to inspect argumentative units and structure, but its moderate performance and the nature of the task support its use as an **inspection aid** [Paper V]. The subsequent collaborative diary-agent work explores a complementary direction in which generated material is not necessarily treated as a completed system interpretation but can become an artefact for interaction, questioning, correction, and revision. This study points towards a broader conclusion of the thesis: person-tailored support requires both computational structures for representing and reasoning about purposeful human activity and opportunities for people to remain involved in interpreting what system-generated support means in their own lives.

The following sections discuss these contributions in relation to the four research questions. They first examine the representation of purposeful activity, then the architectural and argumentation-based mechanisms for purposeful dialogue, followed by argumentation-guided generative and reflective support, and finally the empirical and design knowledge concerning appropriateness,

interpretation, and participation.

7.1 Representing purposeful activity for person-tailored support

The first contribution of this thesis concerns how information about a person's everyday activities can be represented for person-tailored human-agent support. Across the thesis, person-tailoring is treated as more than adapting interaction to demographic characteristics or observed behaviour. Everyday activities are purposeful: they take place in particular contexts, involve people and instruments, and may be connected to goals, motives, preferences, and priorities [59, 49]. Consequently, knowing that an activity has occurred does not by itself provide sufficient information for reasoning about what that activity means to a particular person or how an agent should respond to it.

A central step towards addressing this problem was the development of computer-interpretable representations that connect observations and events with purposeful activity. Paper III provides the main representational contribution by modelling activities together with associated events, actors, instruments, goals, and motives. The model was intended to enable information originating from different sources to be interpreted in relation to the activity in which it occurs, rather than treating observations as isolated data. This provides a basis for distinguishing between an observation of behaviour and an interpretation of that behaviour in relation to purposeful human activity. However, the model was not evaluated with end users and therefore does not demonstrate that such representation by itself improves dialogue, personalisation, or health outcomes.

This representation of purposeful activity also raises the question of what happens when an activity is meaningful for more than one reason. Paper IV addresses this problem by extending the user model towards a multipurpose motivational model. The model represents different motives associated with activities and includes information such as their importance to the person, satisfaction with current performance, and whether support is desired. These motives may reinforce one another, but they may also create tensions. For example, an activity may contribute simultaneously to physical health, social participation, enjoyment, or autonomy, while other circumstances make the same activity difficult or undesirable. Person-tailored support therefore cannot necessarily be organised around a single predefined health objective.

The multipurpose model provides a way of making these different reasons available for computational reasoning. Paper IV further translates the motivational model into a goal model for a team of agents with complementary responsibilities and develops a method for generating arguments that can draw on several motives. In this way, the representation is not only a description of the person but becomes a resource for deciding what reasons may be relevant when providing support. Nevertheless, Paper IV primarily proposes and

implements a model and prototype mechanisms. Its examples illustrate how multipurpose arguments can be generated, but the work does not establish through a user evaluation that the proposed multipurpose modelling leads to more appropriate or effective support [Paper IV].

Taken together, these contributions suggest that person-tailoring requires a distinction between **recording activity** and **representing purposeful activity**. An observation such as walking, resting, meeting another person, or completing an everyday task can acquire different significance depending on the person's goals, motives, circumstances, and priorities. The thesis therefore treats these relationships as part of the computational basis for person-tailored support rather than assuming that the meaning of an observed activity can be inferred from the activity category alone.

This representational perspective also provides a foundation for the later developments in the thesis. The structured representation of activities and motives is reused and adapted when constructing person information for argumentation-based support and, later, for generated narratives. In Paper V, for example, structured activity and motivation profiles provide the persona information against which generated person-specific statements can be assessed. Importantly, the study distinguishes persona grounding from argumentative structure: a statement may be supported by structured persona information without expressing an explicit argumentative premise, while an explicit premise does not itself establish that the content is grounded in the persona [Paper V]. This distinction reinforces the broader thesis argument that representing information about a person and reasoning with that information are related but separate problems.

There are also important limits to this contribution. The representations developed in the thesis capture selected aspects of purposeful activity; they are not comprehensive computational models of a person or of the meaning of human activity. Goals and motives are themselves situated and may change over time as Activity Theory emphasises, and information represented by a system cannot be assumed to capture how a person currently understands an activity. The later studies further show that correspondence with structured information does not guarantee that generated support will be interpreted as relevant or appropriate. The representational contribution should therefore be understood as providing a structured basis for person-tailoring, rather than as establishing a complete or objectively correct representation of personal meaning.

Answer to RQ1. The thesis addresses RQ1 by developing computer-interpretable representations that relate activities and events to selected person-specific information, including context, goals, motives, preferences, and priorities, and by extending these representations to accommodate multiple motives associated with activity. The resulting models provide a computational basis for reasoning about purposeful activity in person-tailored support. The contribution is representational and conceptual, supported by implemented models and prototype

mechanisms; it does not establish that the representations completely capture personal meaning or that their use produces improved health or behavioural outcomes.

7.2 Architectures and argumentation-based mechanisms for purposeful human-agent dialogue

The second contribution of this thesis concerns how structured information about a person, activities, and domain knowledge can be used in purposeful human-agent interaction. Representing purposeful activity provides information that may be relevant to person-tailoring, but representation alone does not determine how an agent should use that information during interaction. The agent also needs to manage the purpose of the dialogue, respond to its unfolding context, and select reasoning and dialogue actions that are appropriate to the person and situation. The thesis addresses this problem through cognitive-agent architecture, purpose-specific dialogue mechanisms, and argumentation-based reasoning.

An early architectural contribution is the cognitive architecture developed in Paper I. The architecture separates components for maintaining beliefs, managing dialogue, managing tasks, and generating plans. This separation reflects the view that human-agent interaction involves more than producing a response to the immediately preceding utterance. The agent needs to relate information obtained during interaction to its knowledge about the person and domain, while also maintaining goals and deciding how the dialogue should proceed. Importantly, the work distinguishes dialogue purposes based on argumentation dialogue types identified by Walton and Krabbe [98] such as information seeking, inquiry, deliberation, and persuasion. The purpose of an interaction therefore becomes part of the computational organisation of the dialogue rather than being treated as an implicit property of its topic.

This architectural contribution should, however, be interpreted according to its implementation maturity. Paper I proposes the broader cognitive architecture, while the dialogue manager was only partially implemented. The work therefore establishes an architectural approach for connecting knowledge, dialogue purpose, and agent reasoning, rather than demonstrating a fully implemented autonomous agent capable of managing all of these processes. This distinction is important because the later papers develop selected mechanisms from this architecture without constituting a complete implementation of all its components.

Paper II provides formative evidence for why managing dialogue purpose and context matters in interaction. The implemented prototype enabled a person to engage with a software agent about health-related issues and was evaluated with five therapists and eleven older adults. The study examined the overall idea and purposes of supportive dialogues, topics of interest, interaction design, and participants' assessments of the appropriateness of individual

agent moves [Paper II]. The study was explicitly formative, and its small sample means that the results should be interpreted as indicative rather than as evidence of general effectiveness.

A useful distinction emerging from this study is between **topic-related appropriateness** and **context-related appropriateness**. A question may concern a topic that is generally relevant to the dialogue while still being perceived as inappropriate at the particular point at which it occurs. Conversely, people's assessments of topic relevance can differ according to how they relate the question to their own circumstances. The prototype therefore exposed a limitation of selecting dialogue content based only on domain relationships: purposeful dialogue also requires sensitivity to the immediate dialogue state and to the person's interpretation of what is appropriate to discuss [Paper II].

The thesis subsequently develops this problem from selecting appropriate dialogue moves towards reasoning about **why particular actions may matter to a person**. Paper IV is central to this development. It starts from the observation that digital coaching is commonly organised around a particular purpose, whereas human activities can be associated with several motives that may reinforce or conflict with one another. The paper therefore extends the motivational user model into a multipurpose model and relates it to a team of agents with complementary goals and responsibilities. A Companion Agent is given a coordinating role intended to maintain a more holistic perspective on the person's situation, needs, preferences, and prioritised activities [Paper IV].

Argumentation provides the mechanism through which these represented motives can become reasons offered during interaction [99]. Paper IV implements different forms of argumentation-based dialogue in NejaVis, including inquiry, deliberation, and persuasion. These dialogue types have different goals: inquiry concerns drawing conclusions about a situation, deliberation concerns deciding what action to take, and persuasion concerns changing a priority or belief. Arguments can be constructed from information available in the user and domain models, and the human can accept or reject claims made by the agent [Paper IV]. Argumentation is therefore not introduced in the thesis only as a method for persuading people. It provides a structured means of connecting **dialogue purpose, claims, reasons, and person-specific information**.

The multipurpose extension is particularly significant for the thesis-level argument. Instead of assuming that an agent should promote an activity because it serves one predefined health goal, the approach allows arguments to draw on different motives associated with the person. Complementary agents represent different domains of concern, while the Companion Agent is intended to mediate the resulting support. This provides a computational approach for reasoning about the fact that the same activity may matter for physical health, social participation, enjoyment, autonomy, or other person-specific reasons. The contribution is therefore not merely the addition of **multiple agents, but the use of multiple motives as resources for constructing person-tailored arguments**.

At the same time, the evidence does not establish that this approach suc-

cessfully resolves conflicting motives in actual use. Paper IV develops and implements the multipurpose motivational model and argument-generation method and illustrates their operation using a use case from earlier work, while identifying user studies to validate the model as future work [Paper IV]. The complementary-agent architecture and orchestration should consequently be understood as a proposed approach to organising multipurpose support rather than as an empirically demonstrated solution to conflicts among human goals. The broader idea of combining specialised computational components is also reflected in later approaches such as Mixture of Experts architectures, although these differ from the proposed complementary-agent approach in their mechanisms and objectives [85].

Across these studies, the role of argumentation also changes. The earlier dialogue architecture establishes that interactions can have different purposes. The formative evaluation shows that appropriate interaction depends on both topic and unfolding context. The multipurpose approach then provides explicit structures for connecting dialogue purposes with reasons derived from a person's activities, priorities, and motives. Together, these developments support a thesis-level view of purposeful human-agent dialogue in which **what the agent discusses, why it discusses it, and what reasons it provides are interconnected aspects of person-tailoring**.

This contribution also provides the conceptual bridge to the later generative work. Argumentation is initially used within structured dialogues to support inquiry, deliberation, and persuasion. Paper V subsequently carries these dialogue purposes and argumentation schemes into LLM-based narrative generation, where argumentation guides how reasons are organised within person-tailored stories. The later results show that this extension can influence explicit argumentative structure without guaranteeing that generated person-specific statements are grounded [Paper V]. This distinction becomes important for the third contribution of the thesis: once reasoning structures are combined with generative models, the problem expands from deciding what arguments to construct to examining what the generated system has actually expressed.

Several limitations therefore delimit the second contribution. The cognitive architecture was not fully implemented; the dialogue evaluation was small and formative; and the multipurpose model and complementary-agent approach were not validated as an intervention with older adults. The thesis consequently does not demonstrate autonomous resolution of competing human motives, sustained adaptation, behaviour change, or health effects. Rather, it contributes computational structures and prototype mechanisms for making **dialogue purpose, context, reasons, and multiple motives explicit components of person-tailored human-agent interaction**.

Answer to RQ2. The thesis addresses RQ2 by proposing an architecture and developing dialogue and argumentation-based mechanisms through which structured person and domain knowledge can be used in purposeful human-agent interaction. Different dialogue purposes provide goals for the interac-

tion, contextual appropriateness informs how dialogue moves need to relate to the unfolding situation, and argumentation provides a mechanism for connecting claims and proposed actions to person-specific reasons. The multipurpose extension further enables several motives to be represented and used when constructing person-tailored arguments. These contributions are supported by architectural models, implemented dialogue mechanisms, prototypes, and formative evaluation, but they do not demonstrate a fully autonomous or empirically validated system for resolving multiple human motives.

7.3 Argumentation-guided generative support, reflective inspection, and collaborative revision

The third contribution of this thesis concerns how the structured and argumentation -based approaches developed in the earlier work can be extended to generative AI while retaining attention to purpose, person-specific information, and human agency. Generative models make it possible to express information about activities, goals, and motives in more flexible narrative forms. At the same time, generation introduces a different problem: even when the system is provided with structured information and an intended interaction purpose, the generated narrative may introduce unsupported person-specific content [46], express reasoning differently from what was intended, or be interpreted differently by the person. The thesis addresses this problem through argumentation-guided narrative generation, separate assessment of grounding and argumentative structure, reflective inspection through argument mining, and an exploratory move towards collaborative interpretation and revision.

Paper V provides the main technical and empirical contribution to the first three parts of this development. The reflective storytelling architecture combines structured user and activity information with dialogue purposes, argumentation schemes, and LLM-based narrative generation. Inquiry, deliberation, and persuasion are operationalised as different communicative purposes, with argumentation schemes used to guide the reasoning expressed in the generated narratives. The approach therefore extends argumentation beyond the structured human-agent dialogues developed earlier in the thesis: rather than using arguments only as explicit dialogue moves, argumentation becomes guidance for organising reasons within generated narrative content [Paper V].

This extension is important because the purpose of a generated narrative is not treated simply as a topic or stylistic instruction. Different dialogue purposes call for different forms of reasoning. Previous work similarly shows that how a story functions as an argument depends on the narrator's intention and dialogical context [12], and that the perceived persuasive properties of argumentation schemes can be influenced by their natural-language instantiation, including topic and stance [82]. A persuasive narrative can provide reasons

connected to values and motives; a deliberative narrative can relate conditions and possible actions; and an inquiry-oriented narrative can support interpretation of past activities or circumstances. Research on narrative persuasion has also shown that the arguments embedded within stories can influence their persuasive effects, although these effects depend on characteristics of both the narrative and its recipients [84]. Argumentation schemes provide structured patterns for these purposes while the language model provides flexibility in how the resulting content is expressed. In this sense, the approach combines symbolic structures that specify aspects of **why and how to reason** with generative mechanisms that determine **how that reasoning is expressed as a narrative**.

The evaluation in Paper V, however, shows that argumentation guidance and grounding need to be treated as distinct properties. In the controlled comparison, the full condition included structured persona and activity information, grounding instructions, dialogue-purpose guidance, and argumentation-scheme guidance. The scheme-ablation condition retained the same persona information, grounding instructions, and dialogue-purpose guidance but removed the argumentation schemes. The standard baseline did not receive the structured persona information, grounding instructions, dialogue-purpose guidance, or schemes [Paper V]. This experimental structure makes it possible to distinguish what can reasonably be attributed to scheme guidance from what belongs to the structured architecture as a whole.

The results show that argumentation-scheme guidance substantially affected **argumentative organisation**. Narratives generated with scheme guidance contained substantially more explicit premises, while the number of claims remained essentially unchanged. This indicates that the schemes influenced how reasons were made explicit rather than simply increasing the amount of assertive content. In this respect, argumentation provides a mechanism for shaping the reasoning structure of generated narratives [Paper V].

The grounding results require a more cautious interpretation. Among sampled applicable statements, 67.6% in the full condition were grounded in the corresponding persona information, compared with 57.9% in the scheme-ablation condition. This difference was not statistically significant. Consequently, the study does not demonstrate that adding argumentation schemes independently improves persona grounding. The full structured condition showed substantially greater sampled persona grounding than the standard baseline, but this comparison concerns the structured person-tailored architecture as a package rather than isolating the effect of argumentation schemes and expected as person-specific information was removed [Paper V].

This distinction between **grounding** and **argumentative structure** is itself an important thesis-level result. Grounding in person-specific information concerns whether a person-specific statement is supported by information in the user model, containing the person-specific information. Argumentative structure concerns whether claims and reasons are made explicit within the generated narrative. A narrative may contain an explicit premise that is un-

supported by the person's represented information, while a grounded statement may occur without being expressed as part of an explicit argument. The two properties are therefore complementary rather than interchangeable [Paper V]. For person-tailored generative support, making reasoning more explicit cannot by itself guarantee that the reasoning is based on appropriate information about the person.

Paper V also investigates whether generated argumentative content can be inspected after generation. Argument mining is used to identify argumentative units and classify them as claims, premises, or non-argumentative content. This reflective analytical layer provides a way to expose aspects of the structure produced by the generative model. However, the evaluation shows clear limits to what this mechanism can establish. The zero-shot argument miner achieved moderate performance against the researcher-agreed consensus annotation and showed systematic classification errors. It should therefore be understood as an **inspection aid**, not as an autonomous verifier [Paper V].

More fundamentally, argument mining answers a different question from factual or persona grounding. Identifying a sentence as a premise does not determine whether the premise is true or supported by the person's information. Likewise, identifying a claim does not establish that the claim is appropriate for the person. Paper V therefore evaluates persona grounding separately against structured persona information. This separation prevents the reflective mechanism from being interpreted as a hallucination detector, factual verifier, or safety guarantee. Instead, its contribution lies in making selected properties of generated argumentative structure available for further inspection.

The older-adult evaluation in Paper V adds another limitation to purely system-centred notions of purpose. The narratives were generated with explicit predefined purposes, but participants did not consistently interpret them in those terms. Across 330 narrative-purpose assessments, an intended purpose was recognised in 30.6%, while 33.3% were assessed as having no clear purpose and 36.1% as expressing another purpose [Paper V]. Thus, encoding an intended communicative purpose and generating a corresponding argumentative structure do not determine how the resulting narrative will be understood by the person. Purpose in person-tailored interaction has both a system-side formulation and a human-side interpretation. This move towards iterative human-agent interaction also relates to broader accounts of human-AI coevolution, which emphasise that humans and AI systems can continuously influence one another rather than interaction proceeding only from system to user [77].

This limitation motivates the next development in the thesis: moving from inspecting generated artefacts towards allowing people to participate in their interpretation and revision. Paper VI explores this direction through the design of a collaborative diary agent. Rather than treating an AI-generated narrative or diary entry as a finished interpretation, the interaction is conceived as a process in which generated material can become a shared artefact [3] that the person can question, correct, elaborate, and revise. In this way,

human participation provides a complementary response to a limitation that automated inspection alone cannot resolve: structured representations and analytical mechanisms cannot determine on their own whether an interpretation adequately reflects what an activity or experience means to the person.

The evidence from Paper VI needs to be interpreted as **design knowledge rather than demonstrated collaborative effectiveness**. The study used participatory design with occupational-therapy experts and developed a partial prototype. Requirements concerning grounding, explainability, emotional safety, ownership, participation, and adaptive interaction emerged from this process. The work therefore identifies properties that a more genuinely collaborative reflective system should support.

Paper V develops and evaluates argumentation-guided narrative generation, grounding, argumentative structure, and automated reflective inspection. Paper VI explores a related but distinct collaborative-interaction direction in which human questioning and revision become more central. The thesis-level connection between them is conceptual: **automated inspection can expose selected properties of generated content, whereas human participation is needed to address interpretations whose appropriateness depends on personal meaning and lived experience**. The thesis does not empirically demonstrate a complete system combining these mechanisms into one validated inspection-and-revision loop.

Taken together, these studies extend the role of argumentation across the thesis. Argumentation first provides structures for purposeful dialogue and reasoning about activities and multiple motives. It is then used to guide the reasoning expressed in generative narratives. Argument mining subsequently makes aspects of the generated argumentative structure available for reflective inspection. The limits observed in both generation and automated inspection, together with the collaborative requirements identified in the diary-agent work, point towards keeping generated interpretations open to human participation rather than treating structured or generated system output as authoritative.

The resulting contribution is therefore not a claim that argumentation solves the reliability problem of generative AI. Instead, the thesis shows how argumentation can provide **explicit structures for purposeful generation and inspection**, while also exposing the boundaries of what those structures establish. Argumentation schemes can influence how reasons are expressed, but they do not guarantee persona grounding. Argument mining can expose argumentative structure, but it does not verify factual or personal appropriateness. Structured grounding can test correspondence with represented person information, but the representation itself does not exhaust the person's interpretation of an experience. These boundaries make human interpretation and revision an important continuation of the person-tailoring approach.

Answer to RQ3. The thesis addresses RQ3 by developing argumentation-guided generative narratives in which structured person information, dialogue purposes, and argumentation schemes guide LLM generation; by distinguishing

persona grounding from explicit argumentative structure; and by investigating argument mining as a mechanism for reflective inspection of generated content. The results indicate that argumentation-scheme guidance can make premises and reasoning structure more explicit, but does not independently establish improved persona grounding, while argument mining is suitable as an inspection aid rather than an autonomous verifier. The thesis further explores collaborative diary interaction as a direction for allowing people to participate in interpreting and revising generated artefacts. This latter contribution is supported by expert participatory design and a partial prototype rather than by evidence of successful collaborative use with older adults. This direction also relates to broader human-AI alignment research, which argues that differences between human and machine reasoning require mechanisms for collaboration, explanation, realignment, and error correction at the interaction-process level [44].

7.4 Empirical and design knowledge about appropriateness, interpretation, and participation

The fourth contribution of this thesis concerns what the empirical studies reveal about how people assess and interpret person-tailored human-agent support. Across different generations of the work, the studies move from evaluating the appropriateness of dialogue moves, to examining how older adults interpret purpose and personal relevance in generated narratives, and finally to eliciting requirements for collaborative interaction with generated diary content. Although these studies differ substantially in participants, systems, and methods, they point to a common issue: **person-tailoring cannot be determined solely from properties encoded in the system. The person remains involved in determining whether an interaction is meaningful and appropriate**.

Paper II provides early formative evidence for this distinction. The study evaluated a prototype human-agent dialogue system with five therapists and eleven older adults. Among the issues examined were dialogue purpose, topics of interest, interaction design, and the perceived appropriateness of agent moves. The results distinguish between the appropriateness of a question in relation to its **topic** and its appropriateness in the **immediate dialogue context** [Paper II]. This distinction is important because an agent may select content that is semantically related to the current topic without that content necessarily being appropriate at that particular moment in the interaction.

The findings also illustrate that appropriateness is partly dependent on the person's interpretation of the dialogue. Different people can relate the same topic or question differently to their own circumstances. Consequently, formal relationships in domain knowledge are useful for identifying potentially

relevant content but cannot fully determine how that content will be received. For person-tailored dialogue, perceived appropriateness therefore involves both the information represented by the system and the situated interaction in which that information is used.

This issue becomes more pronounced when interaction moves from structured dialogue to generated narratives. Paper V investigates narratives produced with predefined communicative purposes and structured persona information. The study involved 55 older adults who evaluated persona-based narratives across inquiry, deliberation, and persuasion purposes. Importantly, participants selected a fictional persona resembling themselves rather than receiving narratives generated from their own personal histories. The results therefore provide evidence about people's responses to **persona-based person-tailored narratives**, not evidence of personalisation based on each participant's own life data [Paper V].

Narrative theory has long emphasised that narratives are interpretive and context-sensitive: readers make sense of them in relation to intentional states, background knowledge, and their understanding of the narrator's intentions [18]. A central finding concerns the difference between intended purpose and perceived purpose. Across 330 narrative-purpose assessments, an intended purpose was recognised in 30.6%, while 33.3% were categorised as having no clear purpose and 36.1% as expressing another purpose [Paper V]. The result shows that implementing a dialogue purpose in the generation process does not determine how the resulting narrative will be interpreted. The computational purpose represents a goal that the agentseeks to accomplishbased on the representations available to it, but purposeful interaction also depends on whether the person recognises that purpose.

This finding connects back to Paper II. In the earlier structured dialogues, a move could be related to the intended topic while still being experienced differently in its immediate context. In the generative setting, a narrative can similarly be produced according to a formally specified purpose without the reader necessarily recognising that purpose. Across these different interaction forms, there is therefore a recurring distinction between **system-side intention and human-side interpretation**. This distinction is a thesis-level synthesis rather than the result of a single study.

Paper V also broadens the empirical picture beyond purpose recognition. Participants assessed relevance, cultural relatability, perceived inconsistencies, and future-use intention. The results were mixed rather than demonstrating uniformly positive acceptance. More than half of the participants reported that they would not use similar functionality in the future, while cultural relatability showed a descriptive relationship with future-use intention [Paper V]. These findings caution against assuming that a narrative becomes personally valuable merely because it is generated from structured profile information or contains recognisable activities and motives.

The technical grounding evaluation in Paper V reinforces this distinction from another direction. A generated statement can be assessed for corre-

spondence with structured persona information, but this establishes a narrower property than personal appropriateness. Persona grounding asks whether person-specific content is supported by the information available in the model; it does not establish whether the content is meaningful, culturally fitting, helpful, or desirable to the person. Similarly, the presence of explicit argumentative reasons does not establish that those reasons reflect how the person understands an activity [Paper V]. Technical grounding is therefore an important condition for responsible person-tailoring, but it is not equivalent to experienced relevance or appropriateness.

The collaborative diary-agent work in Paper VI approaches this limitation from a design perspective. Rather than asking only how the system can produce a better interpretation, the work considers how the person can participate in shaping the interpretation. Through participatory design with occupational-therapy experts and development of a partial prototype, the study identifies requirements concerning grounding, explainability, participation, ownership, emotional safety, and adaptive interaction. These requirements shift attention from personalisation as something performed for a person towards interaction in which generated material can potentially be examined and revised with the person.

This shift should not, however, be interpreted as evidence that collaborative revision has already solved the problems identified in the earlier studies. Paper VI is based on expert participatory design rather than an evaluation with older adults, and the prototype was partial. The identified properties are principally **design requirements and implications**. They do not establish that older adults experience greater ownership, trust, safety, or benefit when using the proposed interaction. This limitation is particularly important when discussing ownership and emotional safety: the study identifies these as concerns that future systems should address rather than validated effects produced by the prototype.

Viewed together, Papers II, V, and VI therefore provide different kinds of evidence at different stages of the research. Paper II provides formative evidence about topic and contextual appropriateness in structured human-agent dialogue. Paper V provides older-adult evaluations of persona-based generated narratives together with evidence that formally encoded purpose is not necessarily perceived as intended. Paper VI provides expert-derived requirements for making interaction with generated diary content more participatory and revisable. These studies should not be combined as if they constitute one longitudinal evaluation or a progressively validated intervention. Their contribution lies instead in the recurring design problem that becomes visible across them.

That recurring problem is the boundary between **computational person-tailoring and experienced person-tailoring**. Computational mechanisms can represent activities and motives, select dialogue purposes, construct arguments, constrain generation, and assess correspondence with structured information. These mechanisms make person-tailoring more explicit and in-

spectable. However, the studies indicate that they cannot on their own determine whether the resulting interaction fits the person's immediate situation, communicates its intended purpose, reflects culturally and personally meaningful interpretations, or supports a form of interaction the person would actually want to use.

This observation also clarifies the role of human participation in the thesis. Participation is not presented as a mechanism that guarantees correctness [27, 50]. A person may change their mind, interpret an experience differently over time, or disagree with an interpretation for reasons that are not represented computationally. Rather, participation provides a way of avoiding the stronger assumption that the system's representation or generated interpretation should be treated as authoritative. The collaborative direction explored in Paper VI therefore complements the structured grounding and reflective inspection investigated in Paper V: system-side mechanisms can expose and constrain aspects of generated content, while human interaction can provide opportunities to question, correct, elaborate, or reject interpretations that depend on personal meaning.

A further limitation concerns the empirical basis of this contribution. The studies use heterogeneous participant groups and evaluation settings. Paper II is a small formative study with therapists and older adults. Paper V evaluates fictional-persona-based narratives with older adults rather than narratives generated from participants' own histories. Paper VI involves occupational-therapy experts rather than older adults and examines requirements and a partial prototype. The contribution should therefore be understood as **formative empirical and design knowledge about person-tailored interaction**.

Answer to RQ4. The thesis addresses RQ4 by providing formative empirical and design knowledge about how person-tailored human-agent support is assessed and interpreted across structured dialogue, generative narratives, and collaborative diary interaction. The studies indicate that appropriateness depends on more than topic selection, that an encoded communicative purpose is not necessarily perceived as intended, and that correspondence with structured person information does not by itself establish personal relevance or appropriateness. They further identify participation, correction, grounding, explainability, ownership, and emotional safety as considerations for collaborative interaction with generated content. These findings support designing person-tailored systems in which computational representations and reasoning guide support without replacing the person's role in interpreting and potentially revising what that support means.

Chapter 8

Conclusions and Future Work

This thesis investigated how person-tailored human-agent dialogues can be modelled, implemented, and evaluated to support health-related reflection and planning, with a focus on older adults. The work began from the premise that person-tailoring requires more than adapting information to demographic characteristics or recorded behaviour. Everyday activities are undertaken in particular circumstances and may be connected to several goals, motives, priorities, and social meanings. The research is grounded in Activity Theory [34] and Argumentation Theory [99], which provide complementary foundations for understanding purposeful human activity and developing computational models for reasoning and person-tailored dialogue. An agent therefore needs ways to represent purposeful activity, use this knowledge according to the purpose of an interaction, and keep its interpretations open to human examination and revision.

The thesis contributes a knowledge-based and argumentation-oriented approach that connects five related capabilities: representing purposeful activity, organising purposeful dialogue, reasoning with multiple motives, guiding generative narratives, and inspecting and collaboratively interpreting generated content. These capabilities were developed through conceptual models, cognitive-agent architectures, implemented mechanisms and research prototypes, formative evaluations, and participatory design. They do not constitute a completed clinical intervention or a fully validated production system.

The central conclusion is that computational person-tailoring and experienced person-tailoring are related but not equivalent. Structured knowledge can help an agent connect activities with goals, motives, context, and dialogue purposes. Argumentation can make claims, reasons, and interaction purposes more explicit. Generative models can express these relationships in flexible narrative forms. However, none of these mechanisms can independently determine

whether the resulting support is meaningful, appropriate, or useful to a particular person. Person-tailored support therefore requires both computational structures and continued human active participation [41].

8.1 Conclusions in relation to the research questions

RQ1 asked how knowledge about users, activities, goals, motives, dialogue purposes, and context can be represented to support person-tailored human-agent dialogues. The thesis addresses this question by developing computer-interpretable representations that relate activities and events to selected information about the person and the circumstances in which the activity occurs. These representations distinguish the recording of behaviour from the modelling of purposeful activity. They also accommodate the possibility that one activity may serve several motives and that these motives may reinforce or conflict with one another [Paper III, Paper IV].

The resulting models provide a structured basis for person-tailoring. They make selected relationships among activities, goals, motives, priorities, actors, instruments, and context available for computational reasoning. However, they are models of selected aspects of purposeful activity rather than complete representations of personal meaning. A represented motive cannot be assumed to remain stable, and correspondence with stored information does not establish that an interpretation reflects how the person currently understands an experience.

RQ2 asked how cognitive-agent architectures can integrate such knowledge to support collaborative reasoning and decision-making in person-tailored human-agent dialogues. The thesis addresses this question through a cognitive architecture and dialogue mechanisms that connect person and domain knowledge with different dialogue purposes. Information seeking, inquiry, deliberation, and persuasion provide different goals for interaction, while argumentation connects claims and possible actions with person-specific reasons [Paper I, Paper II].

The multipurpose approach extends these mechanisms by representing several motives and using them to construct person-tailored arguments. A team of agents with complementary responsibilities and a coordinating Companion Agent was proposed and prototyped as a way of organising support across different health-related purposes [Paper IV]. This is an architectural and computational contribution. The thesis does not demonstrate that the proposed agent team autonomously resolves competing human motives or improves health-related decision-making in sustained use.

The formative evaluation of structured human-agent dialogue further indicates that selecting a generally relevant topic is not sufficient. A dialogue move may concern an appropriate topic but still be inappropriate at a particular point in the interaction [Paper II]. Purposeful dialogue therefore depends

on the represented knowledge, the intended purpose, the unfolding dialogue context, and the person's interpretation of the interaction.

RQ3 asked how adaptive dialogue agents can support person-tailored health-related reflection and planning through argumentation, storytelling, and reflective inspection of generated content. The thesis addresses this question by extending dialogue purposes and argumentation schemes into LLM-based narrative generation. In the reflective storytelling architecture, structured activity and motivation information provides a basis for person-tailoring, while dialogue purposes and argumentation schemes guide the reasoning expressed in generated narratives [Paper V].

The latest evaluation in Paper V shows that argumentation-scheme guidance primarily influenced explicit argumentative structure. It substantially increased the presence of premises without producing a corresponding increase in claims. The evaluation did not find a statistically significant improvement in sampled persona grounding when the scheme-guided condition was compared with the otherwise equivalent scheme-ablation condition. The full structured architecture showed substantially greater persona grounding than the standard baseline, but that comparison evaluates the combined effect of persona information, grounding instructions, dialogue-purpose guidance, and argumentation schemes rather than isolating any one component [Paper V].

Consequently, argumentative structure and persona grounding must be treated as separate properties. A generated statement may provide an explicit reason without being supported by the represented information about the person. Conversely, a grounded statement may not form part of an explicit argument. Argumentation can structure how reasons are expressed [99], but it does not guarantee factual correctness, personal appropriateness, or grounding.

Argument mining provides a further mechanism for inspecting generated argumentative content. It can make predicted claims, premises, and non-argumentative material available for examination. Its moderate performance and systematic classification errors, however, limit what can be inferred from its output. Argument mining is therefore an inspection aid, not an autonomous verifier, factual checker, hallucination detector, or guarantee of narrative safety [Paper V].

The collaborative diary-agent work develops the human-participation side of RQ3. It explores how generated material could become a shared artefact that a person can question, elaborate, correct, and revise. Participatory design with occupational-therapy experts identified grounding, explainability, emotional safety, ownership, participation, and adaptive interaction as important requirements [2]. These are design findings supported by expert participation and an exploratory, partially implemented prototype. They are not evidence that collaborative revision has been successfully achieved with older adults or that it improves reflection, planning, or health outcomes.

RQ4 asked how domain experts and older adults perceive and experience the support provided by such agents. The studies were exploratory aiming for formative insights, which could guide further development. The early dia-

logue evaluation study showed that therapists and older adults distinguished between the general relevance of a topic and the appropriateness of a move in its immediate dialogue context [Paper II]. This demonstrates why appropriateness cannot be inferred solely from formal relationships in the system's domain knowledge.

Paper V provides broader empirical evidence from domain experts and 55 older adults evaluating persona-based generated narratives. The older adults evaluated narratives based on fictional personas rather than narratives derived from their own activity histories. The findings therefore concern responses to persona-based narratives and should not be interpreted as evaluation of fully personal data-based support.

The results were mixed. The narratives were rated as moderately relevant overall, and participants did not consistently recognise their intended communicative purposes. Across 330 purpose assessments, the intended purpose was recognised in 30.6%, another purpose was selected in 36.1%, and no clear purpose was identified in 33.3%. Many participants also indicated that they would not use similar functionality in the future [Paper V]. These findings show that encoding a purpose in the generation process does not determine how the narrative will be interpreted or whether the functionality will be considered useful.

The participatory diary-agent study complements these findings by identifying requirements for more collaborative forms of interaction with generated content [Paper VI]. Together, the studies indicate that computational mechanisms can constrain and expose aspects of person-tailored support, but people remain involved in determining its relevance, meaning, and appropriateness.

The overall aim of the thesis has therefore been addressed through an integrated investigation of representation, architecture, dialogue, argumentation, narrative generation, reflective inspection, and collaborative design. The thesis shows how selected knowledge about purposeful activity can be made computationally available and used to guide person-tailored interaction. It also establishes important limits: represented knowledge does not exhaust personal meaning; dialogue purpose does not ensure perceived purpose; argumentative structure does not guarantee grounding; technical grounding does not establish experienced relevance; and automated inspection does not replace human judgement.

8.2 Future Work

The first priority for future work is to integrate the contributions into a more complete interactive system. The thesis develops connected models and mechanisms, but it does not empirically evaluate a single system combining purposeful-activity representation, multipurpose reasoning, argumentation-guided generation, automated inspection, and collaborative human revision. Such integration should preserve the distinctions established in this thesis rather than treating

the output of one mechanism as validation of another.

The representations of activities and motives should be made more dynamic. Future systems should allow people to add, correct, reprioritise, and remove information about their activities, goals, motives, and preferences. They should also represent uncertainty and change over time. This would reduce the risk of treating an earlier or system-inferred representation as an authoritative account of the person.

The multipurpose model and complementary-agent approach require empirical evaluation. Future studies should examine whether representing several motives helps people understand alternatives, negotiate competing priorities, or identify activities that serve several personally important purposes. Such studies should evaluate not only whether the system can generate multipurpose arguments, but also whether people consider the represented motives and resulting reasons accurate, appropriate, and useful.

The generative storytelling architecture should be evaluated in interactive use with narratives grounded in participants' own activity information. This is needed because the older-adult study in Paper V used predefined personas and isolated narrative evaluations. Future work should examine how people respond when they recognise their own activities, can correct the underlying information, and can question or revise the resulting narrative, to accomplish this, the user model needs to be maintained transparent to the user and negotiable [27, 50, 28]. Longitudinal studies are also required to investigate sustained use and changes in perceived relevance, trust, ownership, and usefulness.

Further technical work is needed on grounding and inspection. Grounding mechanisms should distinguish clearly between direct support from represented person information, reasonable general information, unsupported personal assumptions, and contradictions. Any automated grounding or argument-mining output should remain visible as uncertain analytical support rather than being presented as verified truth. Improvements in argument mining should address the confusion between rhetorical encouragement, descriptive content, claims, and premises, as well as argumentative units that perform more than one function.

The collaborative diary agent provides a basis for studying human participation in narrative construction. The next stage should complete the prototype and evaluate it with older adults, including people with different levels of digital and cognitive ability. Evaluation should examine whether users can understand the agent's contribution, correct its assumptions, control the purpose and content of diary entries, and retain ownership of the resulting artefact [3]. Emotional safety and the balance between guidance and autonomy should be evaluated as interaction outcomes rather than assumed from the design.

Finally, future research should examine the conditions under which person-tailored dialogue support contributes to reflection, planning, or health-related action. The present thesis provides models, mechanisms, formative evidence, and design requirements. These outcomes require long-term studies with appropriate measures, comparison conditions, and involvement from intended users

and relevant professionals.

In conclusion, person-tailored human-agent support should not be understood as a system independently determining what a person's activities mean or what the person should do. Its more defensible role is to organise relevant knowledge, make purposes and reasons more explicit, generate material for reflection, expose selected properties for inspection, and provide opportunities for the person to question and reshape the resulting interpretation. This combination of computational structure and human participation provides the central foundation for future development of purposeful, person-tailored dialogue agents.

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